Character Modelling

09

Character modelling is the process of creating digital three-dimensional models. These are commonly used in animation, video games, and visual effects. Blender is a popular open-source 3D modelling and animation software that provides a comprehensive set of tools and features for character development.

Character modelling has numerous applications in various industries. Here are some common applications of character modelling:

Animation: Character modelling is extensively used in animation, including films, television shows, and online videos. Characters are created and animated to bring stories and narratives to life.

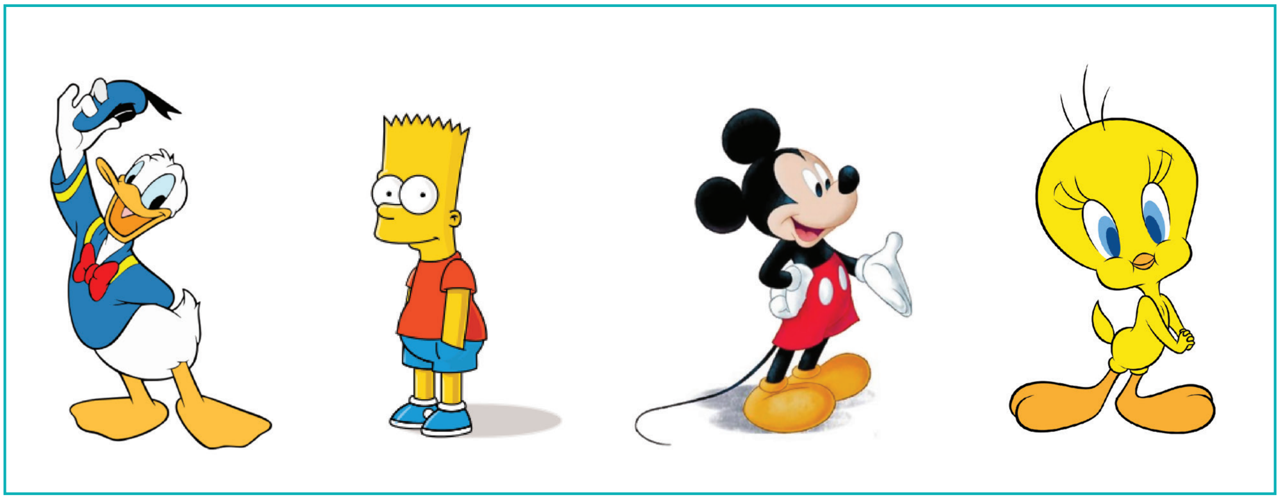


Image Credits: Wikipedia.org

Video Games: Character modelling is essential in video game development. Game designers create complex, realistic, and stylized characters to populate game worlds.

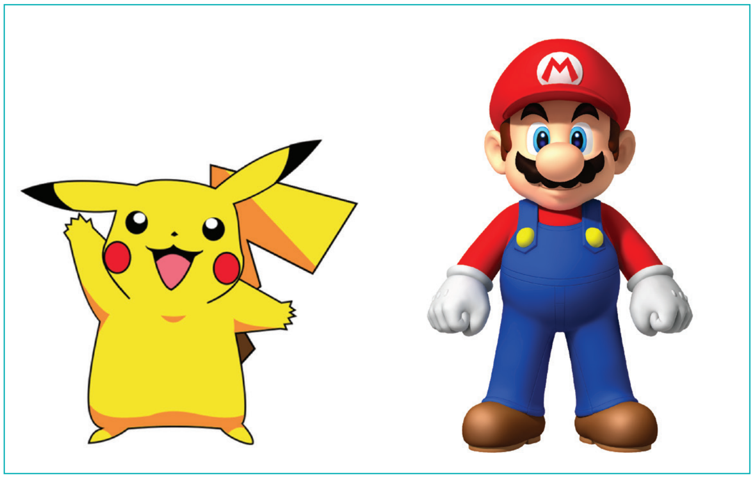


Image Credits: Wikipedia.org

Virtual Reality (VR) and Augmented Reality (AR): Character modelling plays a significant role in VR and AR applications. Characters are created to populate virtual or augmented environments, allowing users to interact with them in real time.

Advertising and Marketing: Advertising companies use character modelling to create mascots and memorable characters to represent their brands, products, or services.

Image Credits: Wikipedia.org, kelloggs.com, duracell.in

Education and Training: Character Modelling is used by educational or training institutions to create characters that simulate realistic scenarios or act as virtual instructors. These provide interactive and engaging learning experiences.

Art and Design: Artists and designers create unique characters for personal projects, concept art, illustrations, comics, and graphic novels.

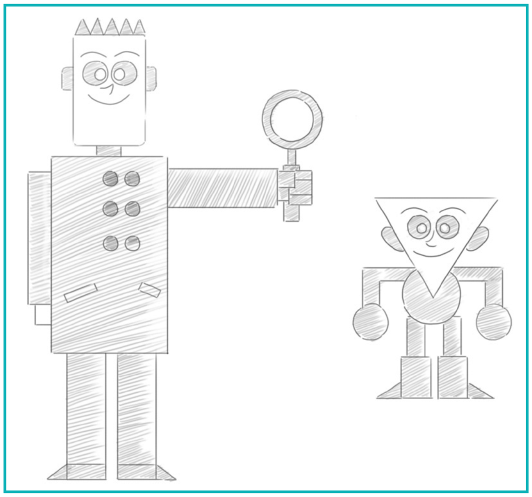
The discussed examples are just a few of the diverse applications of character modelling. The versatility and flexibility of character modelling make it an integral part of various creative industries, entertainment, and educational fields.

Before starting the modelling process, it is essential to have a clear idea of the character's appearance and design. This can be achieved by using **Concept Art** or **Reference Images** that serve as a guide during the modelling process.

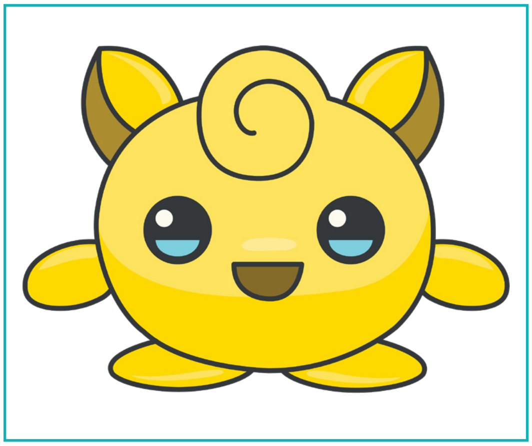
Concept Art

Concept art involves creating visual representations of characters through illustrations, sketches, or digital paintings. It serves as a reference for 3D modellers who convert a 2D concept into a 3D model.

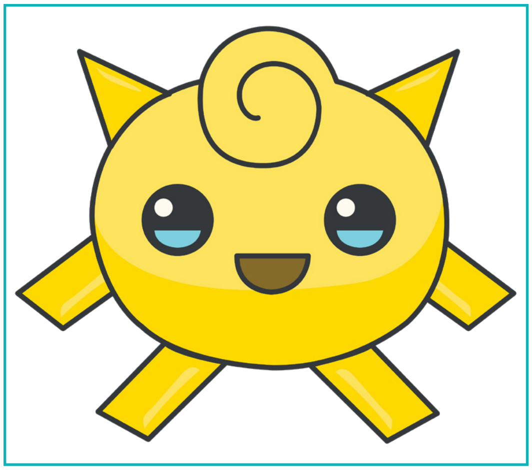
Concept art allows artists to explore different design possibilities for a character. It allows experimentation with shapes, dimensions, costumes, hairstyles, accessories, and other visual elements to find the most appealing and appropriate look. Modellers often start with basic geometric shapes as a foundation. These shapes act as a framework to block out the overall proportions and pose of the character.



In order to understand this better, we will model a simple character that we can create using basic shapes. Let us create a model that is based on the Pokémon character, Jigglypuff.



Can you figure out how we can create this character using basic shapes?

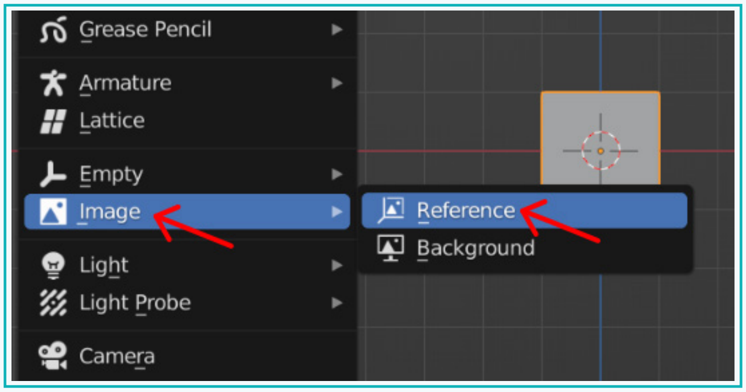


Activity

Set up the Reference Images

We will set up the reference image in the 3D viewport so that we can accurately create the 3D model.

1. Go into the front view. Press ‘Shift + A’ → ‘Image’ → ‘Reference’.

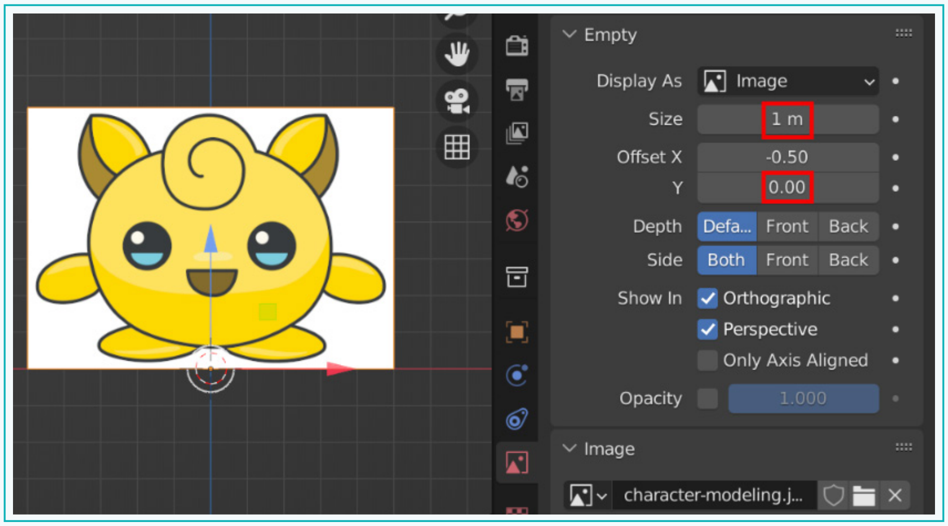


2. Navigate to the location where the reference image is saved and click on the ‘Load Reference Image’ button. Keep ‘Align to View’ enabled.

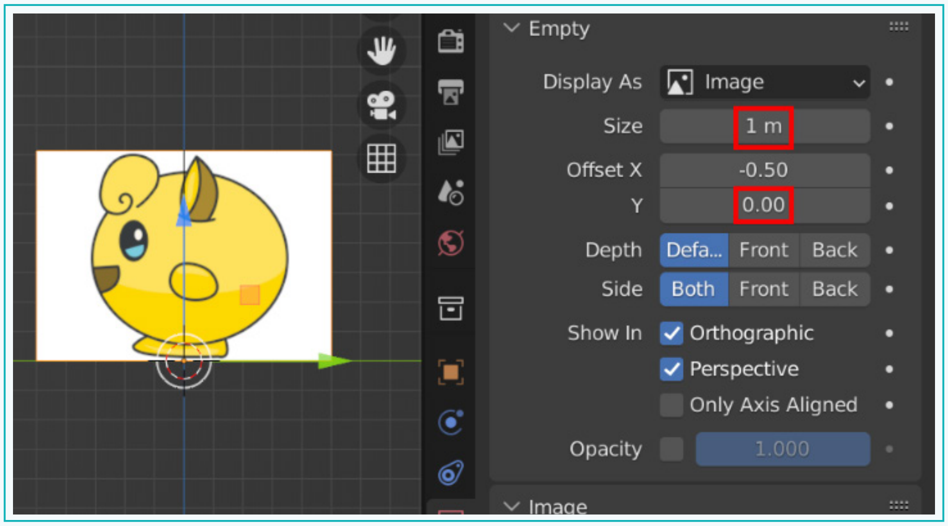
The ‘Align to View’ option aligns the image with the view that the 3D viewport is in. As we are in the front view in the 3D viewport, the image will also be aligned with the front view.

Activity

3. Next, tab into the 'Object Data Properties' window in the 'Properties' editor and reduce the size of the image. Also, change its 'Offset Y' value such that it is standing on the grid.



We will also load a side reference image to determine the appropriate side profile of the character. Follow the same steps to load the side reference image. However, make sure to be in the 'Side view' ('numpad 3') before loading the image.

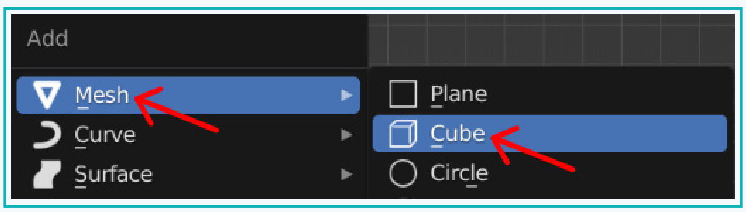


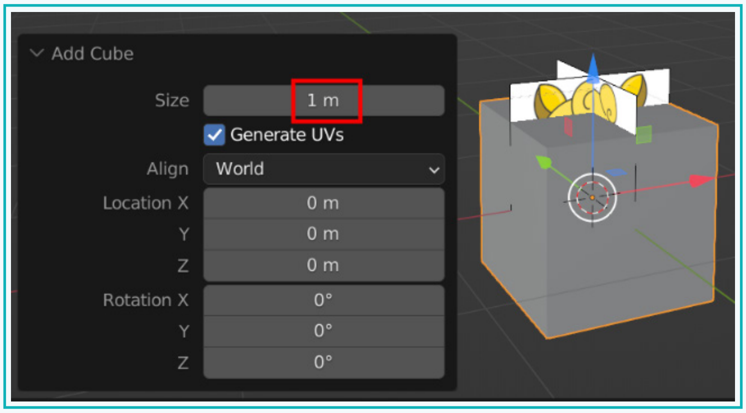
Activity

Start Modelling

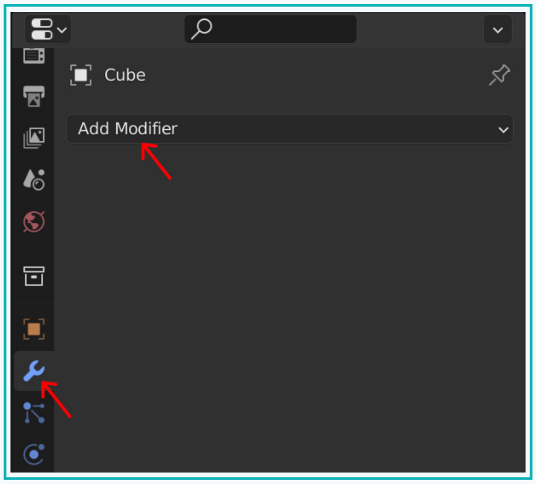
Let us start with a cube.

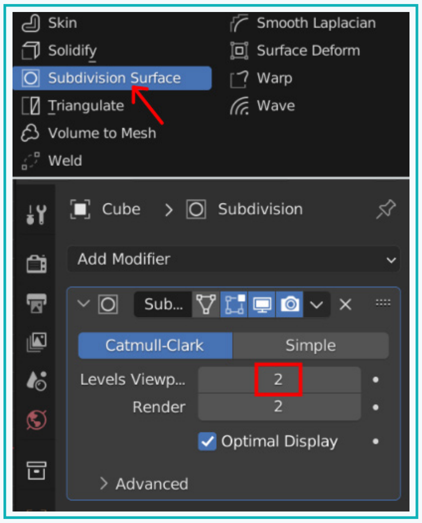
1 Create a cube (press 'Shift + A').



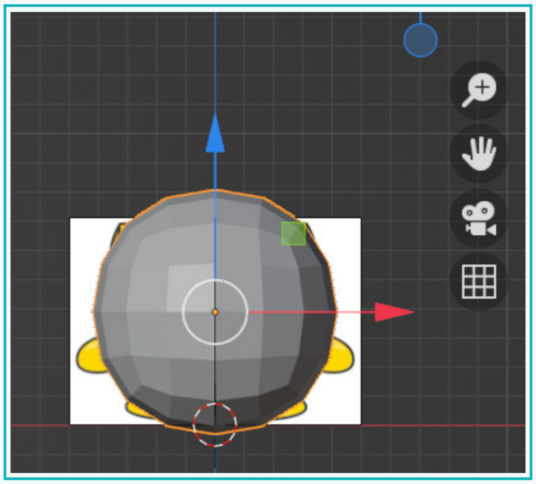


2 Add the 'Subdivision Surface' modifier to the cube.





Activity

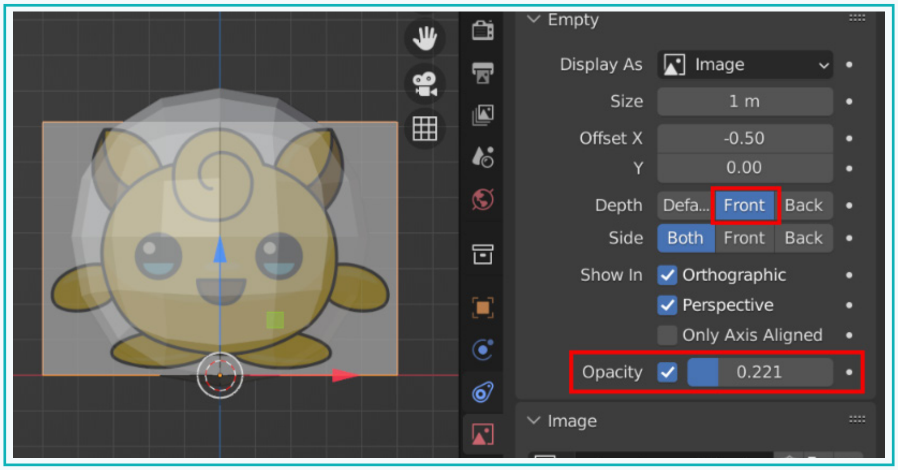


The shape that we have modelled now resembles a UV Sphere. This indicates that we can also use a UV Sphere instead of a cube. Generally, a cube is easier to manipulate. However, the decision of which to use depends on the particular project requirements. Adding a 'subdivision' modifier creates smoother, more rounded surfaces. It also allows for more control over the topology since we start with a grid structure.

On the other hand, in the UV Sphere, the faces are not all the same size. The faces on the top are triangular, while the centre faces are more quadrilateral. Irregular faces create difficulty to choose specific areas and also require extra effort to achieve certain effects.

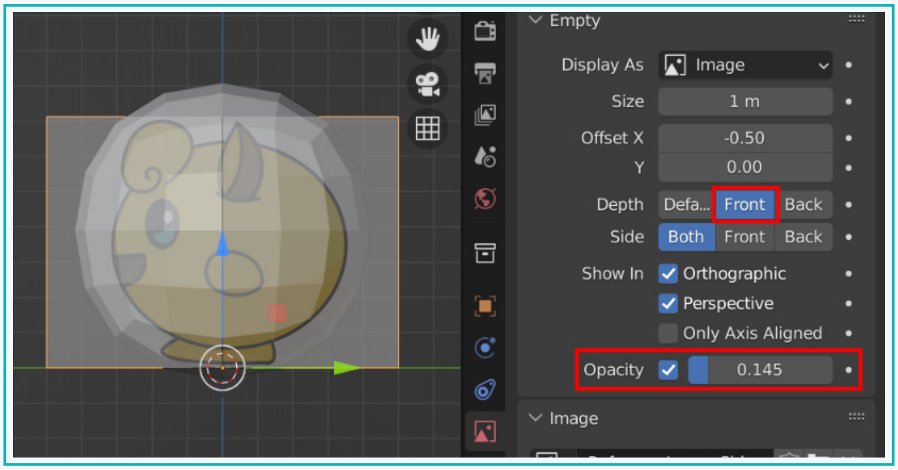
With the reference image now displayed in the viewport, it is important to see both the object and the reference image at the same time. This allows us to model while referencing the background image.

3. We can achieve this by changing the 'Depth' property to the 'Front' in the 'Object Data Properties' window. Also, enable the 'Opacity' property and adjust the opacity value, as required.



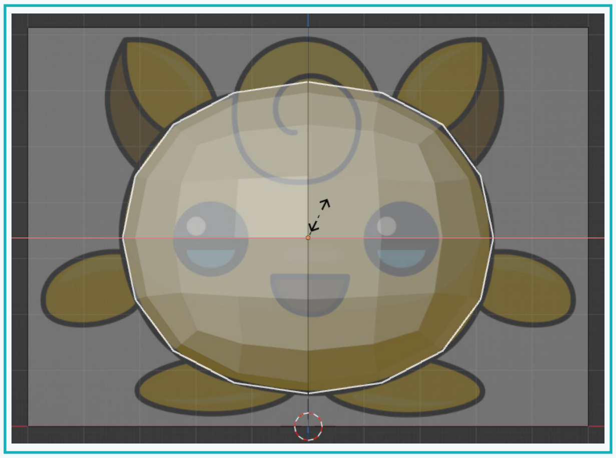
Activity

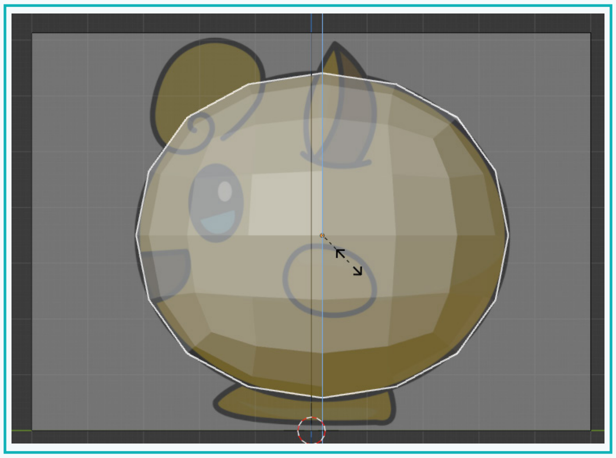
4 Also, repeat the same for the side view.



With the reference image in view, we can now precisely model the character. Scale it to match the reference image.

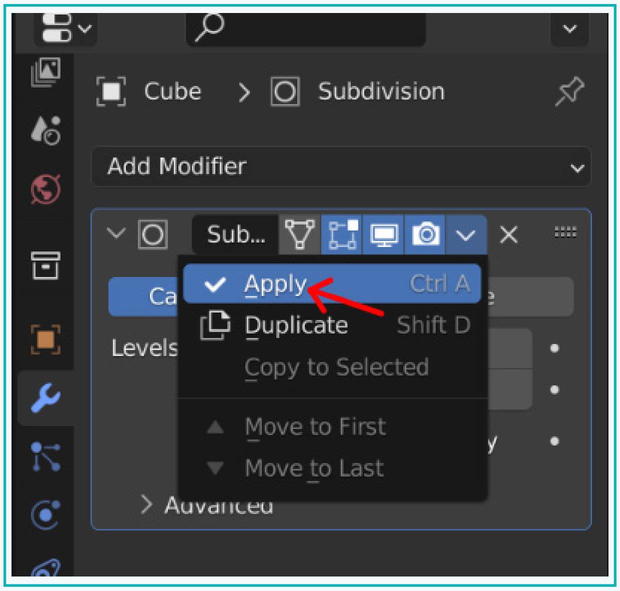
5 Press ‘S’ followed by ‘Z’ to scale the model on the z-axis. Similarly, press ‘S’ followed by ‘X’ to scale on the x-axis. Repeat for the side view as well. For the side view, scale in the z and y axis, as required.



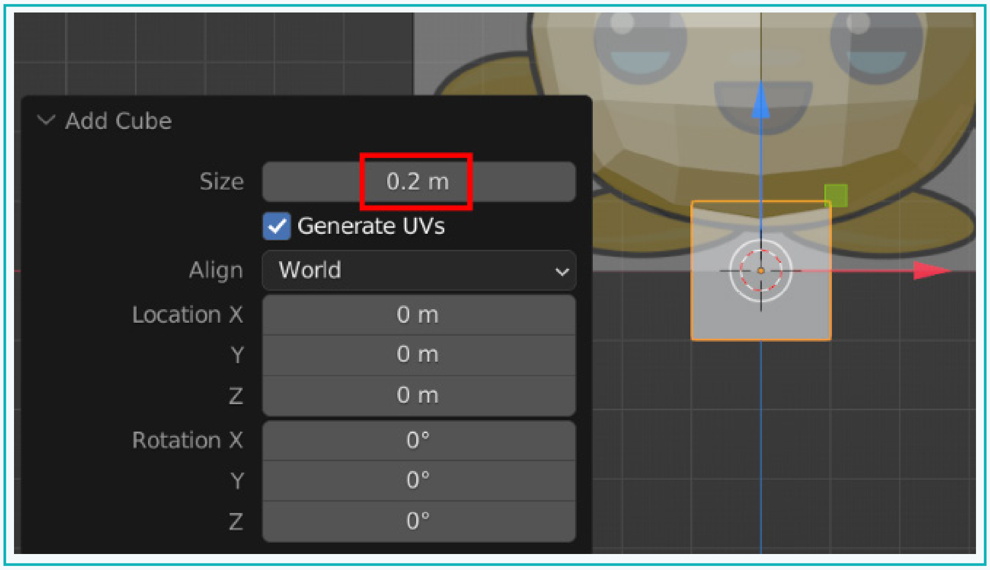


Activity

6 Once done, apply the 'Subdivision' modifier. Applying a modifier allows us to manipulate the faces of the object.

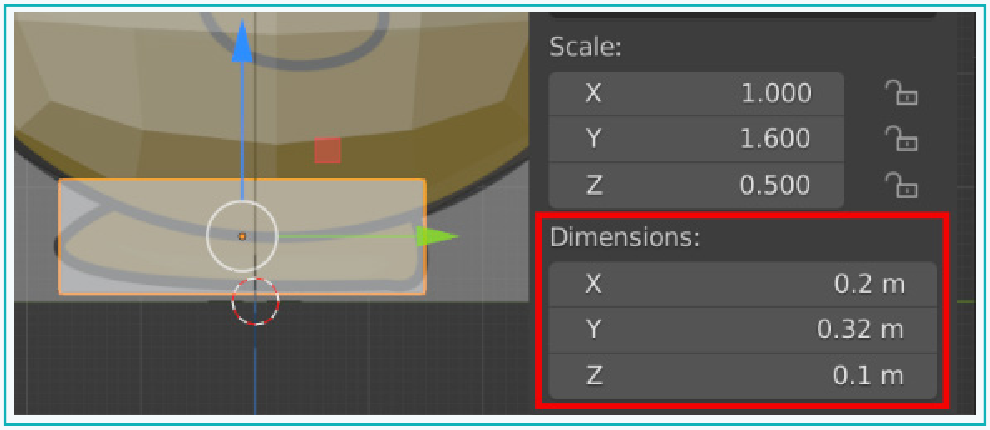


7 Next, let us start with the feet of the character. Create a cube of the given size.

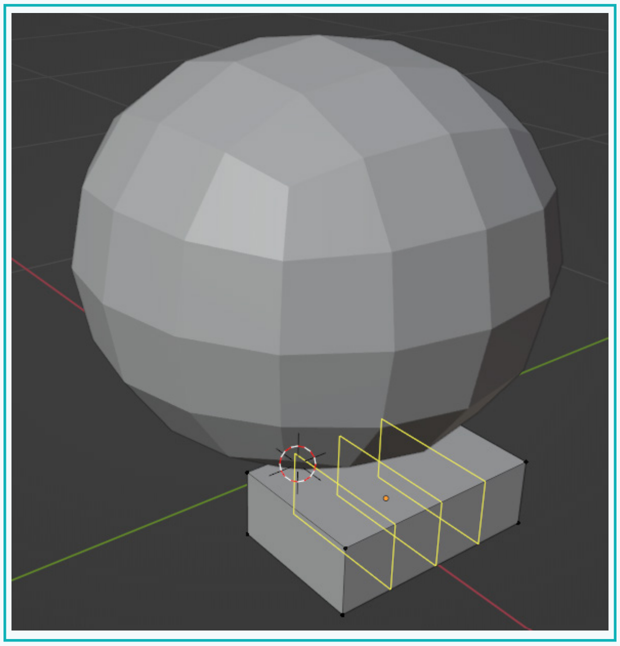


Activity

8. Press 'N' for the 'Transform' panel and change its dimensions.

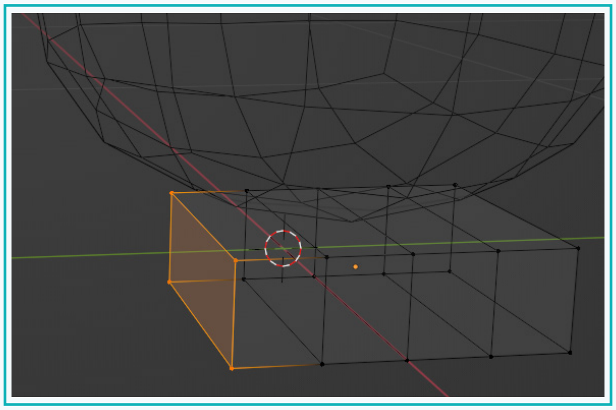


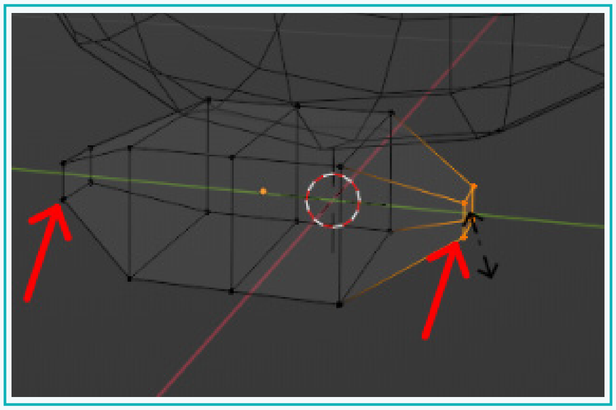
9. Tab to go into the 'Edit' mode. Press 'Ctrl + R' and add three edge loops, as shown.



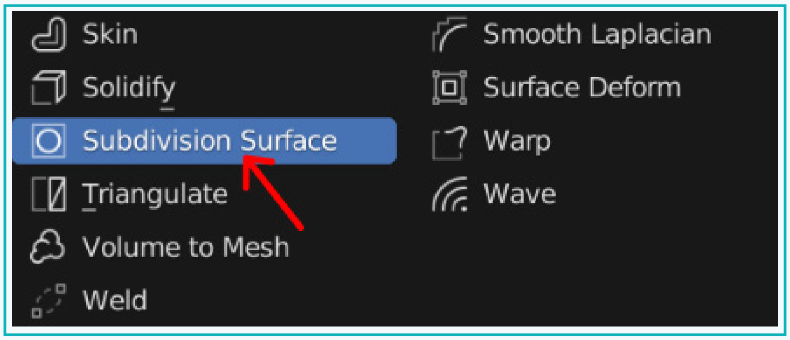
Activity

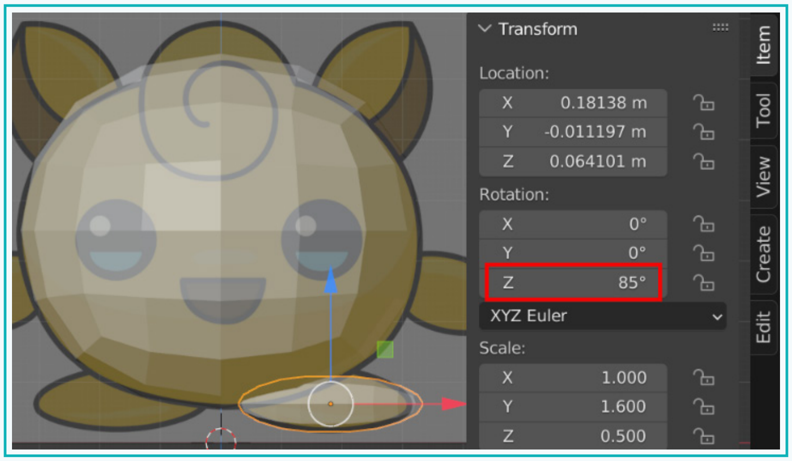
10 Scale down the ends of the cube to obtain tapering shapes on both ends.





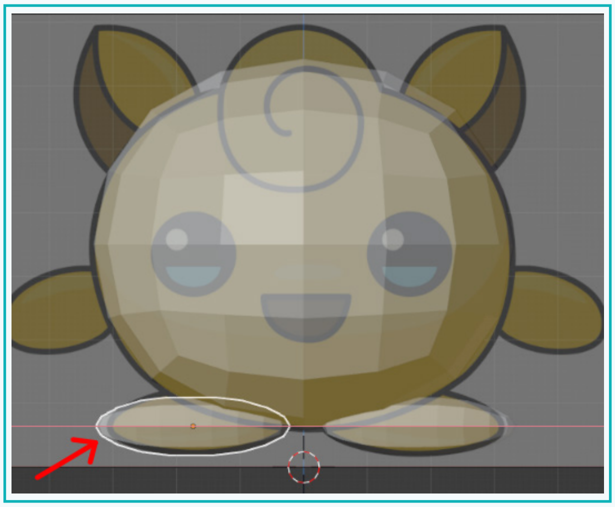
11 Next, add the ‘Subdivision Surface’ modifier and rotate the feet to align them with the reference image.

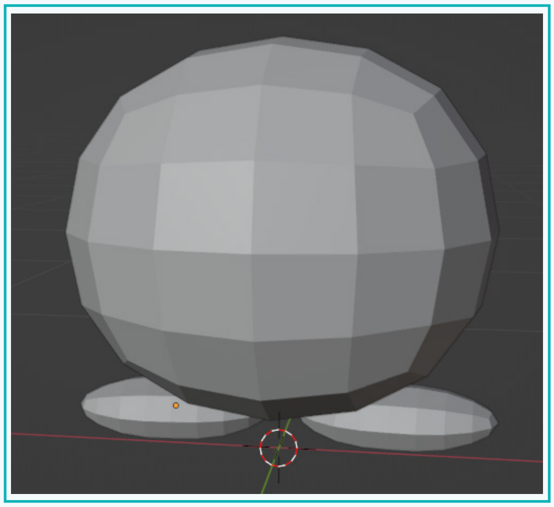




Activity

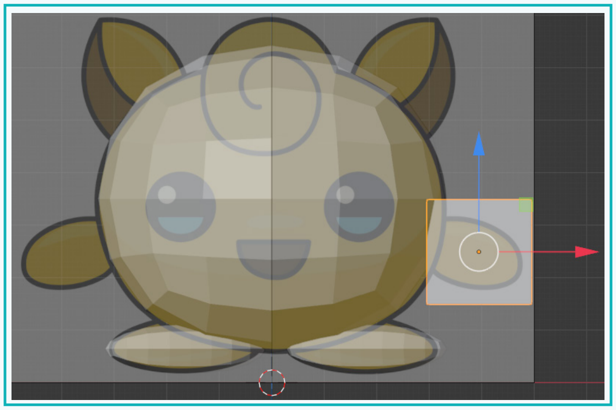
12 Press ‘Shift + D’ (to duplicate), followed by ‘X’. This duplicates the feet on the x-axis. Position as per the reference image.

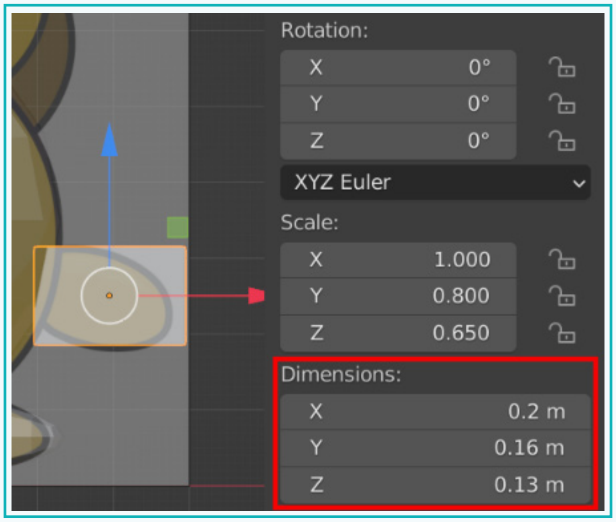




Similarly, we will create the hands of our character.

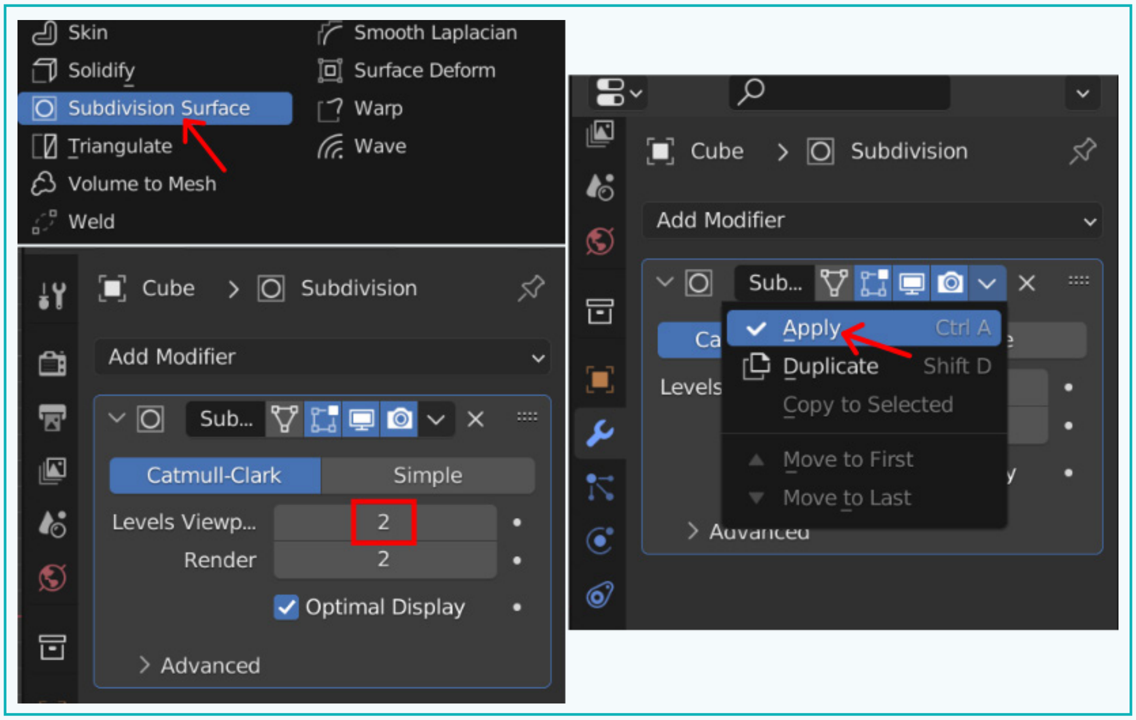
13 Create a cube and position it in place. Change the dimensions, as shown.



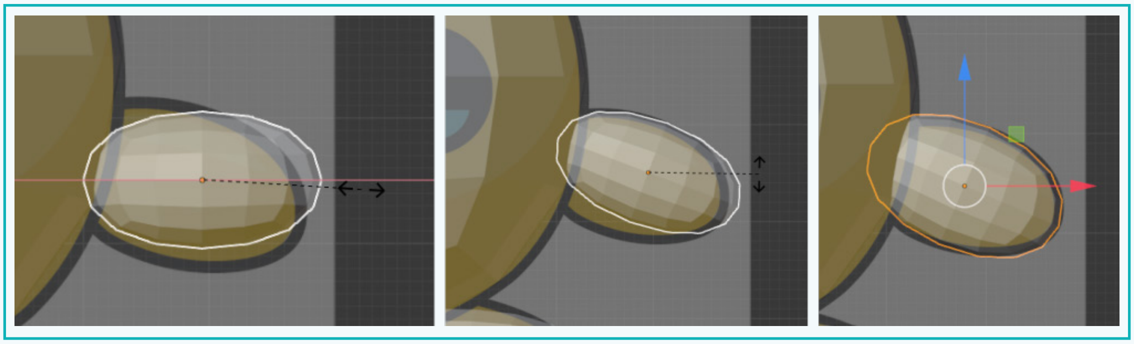


Activity

14 Add the 'Subdivision Surface' modifier and apply it.

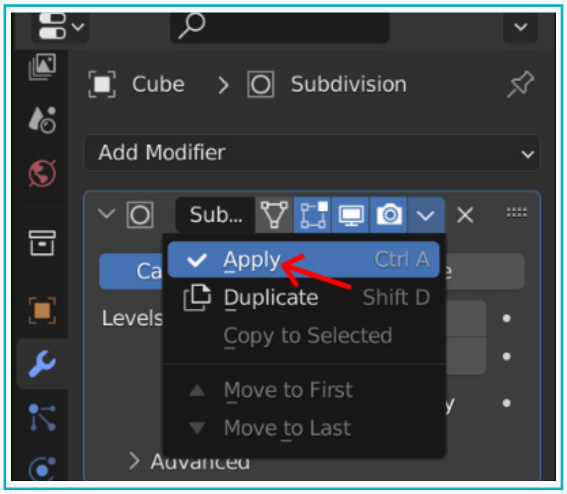


15 Resize and position to match the reference image.

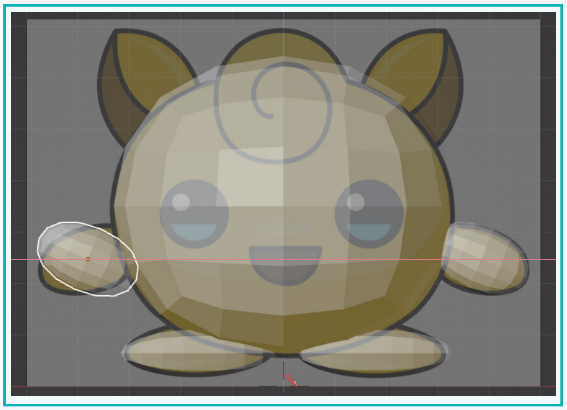


Activity

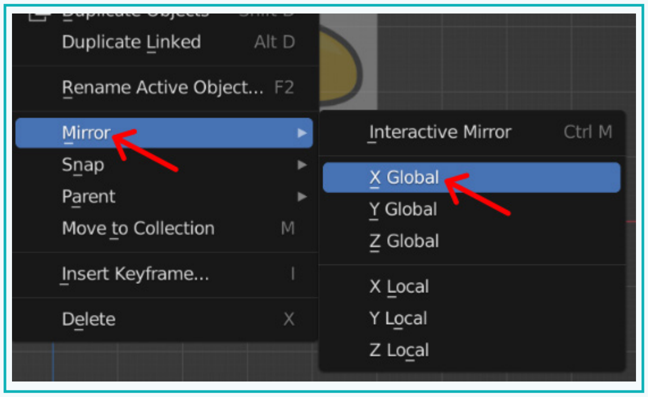
15 Apply the modifier.

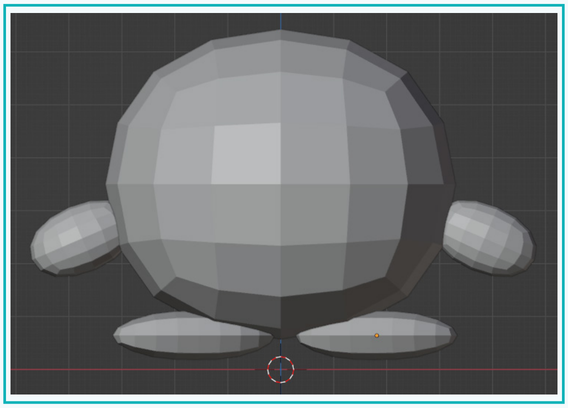


16 Duplicate to create the other hand. Press 'Shift + D' followed by 'X'.



17 Mirror its position by right-clicking over it and selecting 'Mirror' → 'X Global'.

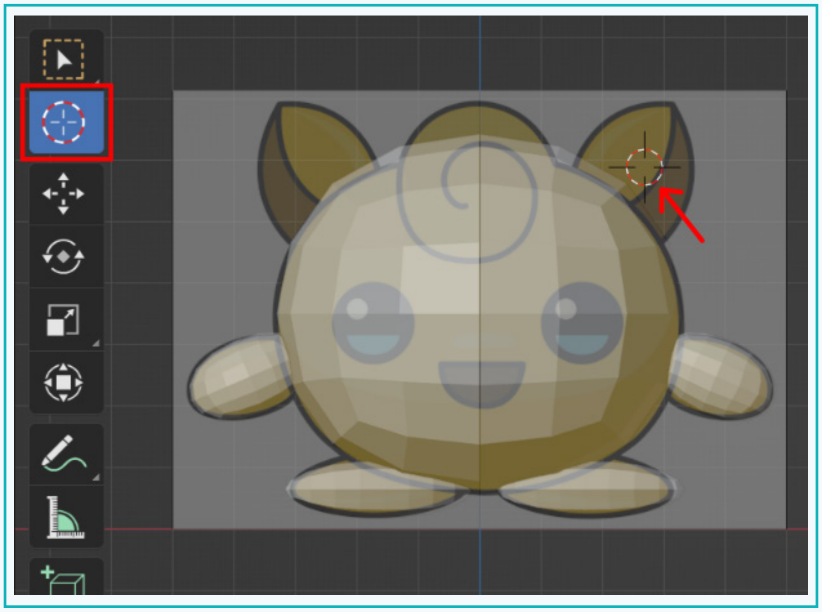




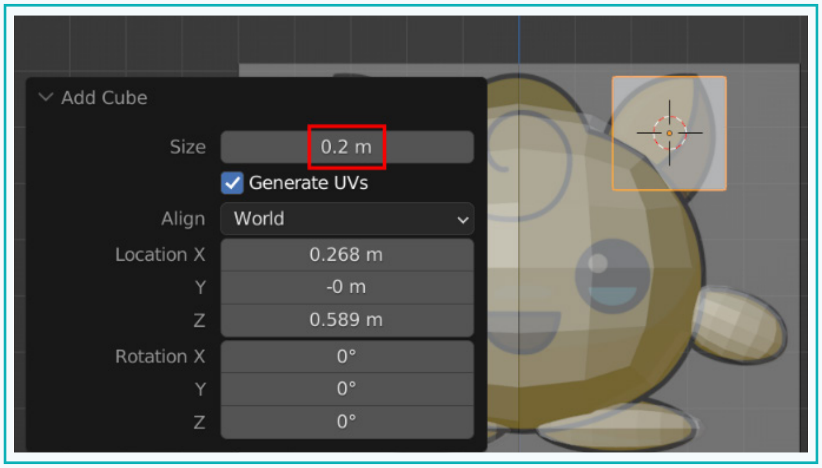
Activity

The next part of the character that we will create is, ears. We will use the 3D cursor to simplify the positioning.

18 Either press 'Shift + Right-click' at the appropriate position where you want the 3D cursor to be in, **OR** click on the 3D cursor tool in the toolbar and left-click at the desired position.

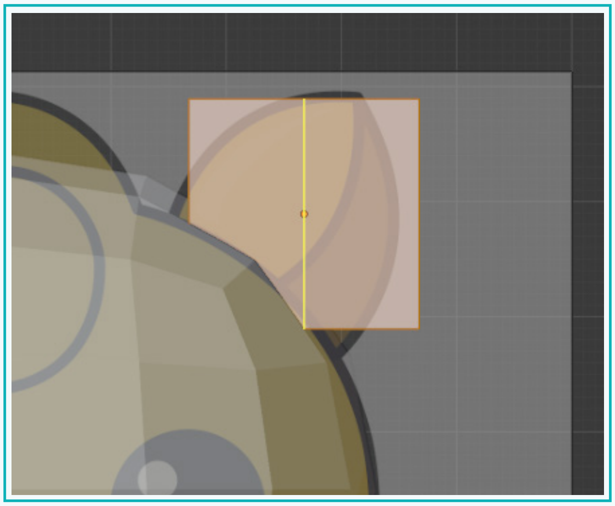


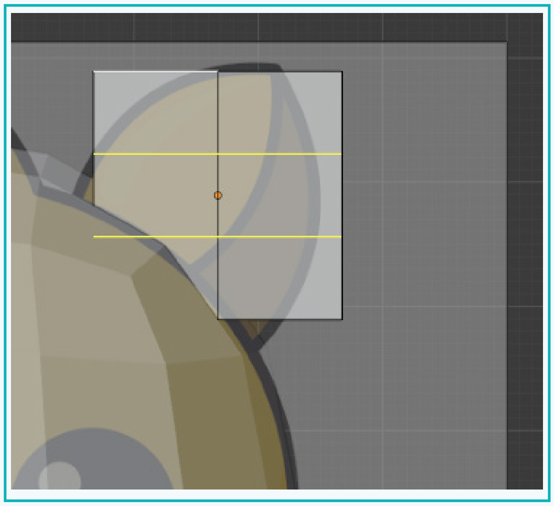
19 Now create a cube with the given dimensions. It gets created at the position of the 3D cursor, thus eliminating the need to manually position the cube.



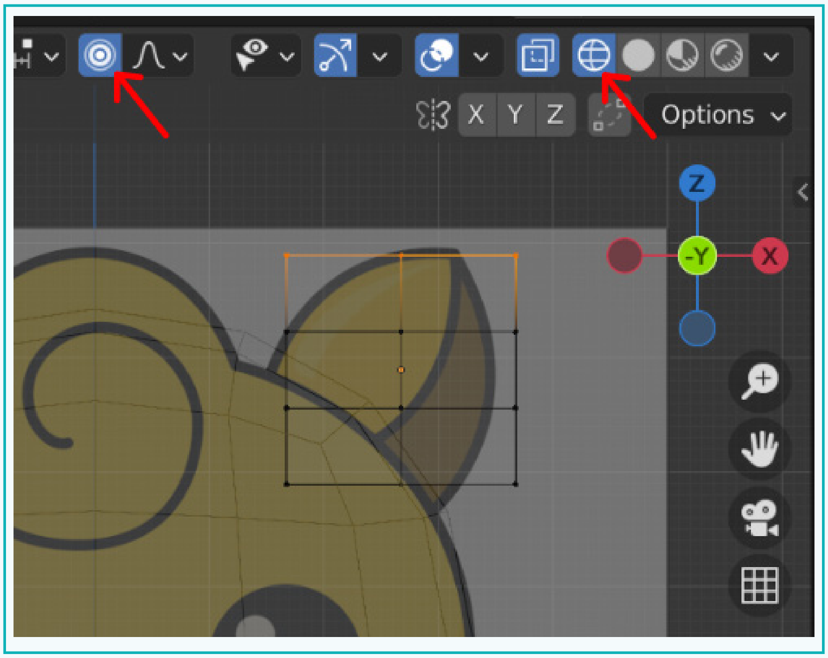
Activity

20 Tab into the 'Edit' mode and press 'Ctrl + R' to add one vertical and two horizontal edge loops.





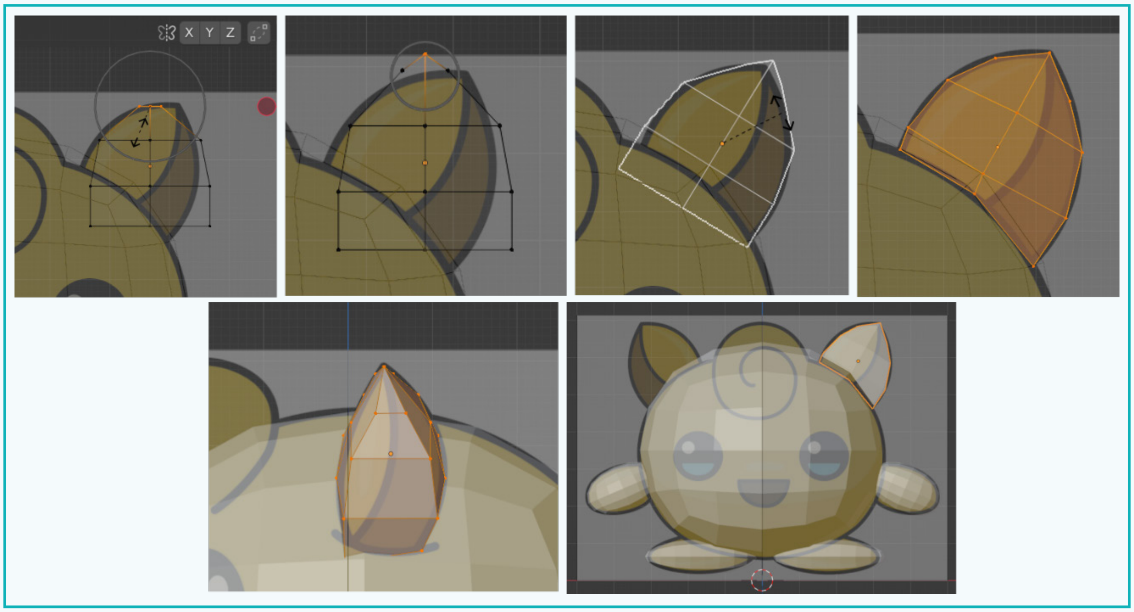
21 Go into the 'Wireframe' mode and switch on 'proportional editing'. Select the top rows of the vertices, as shown.



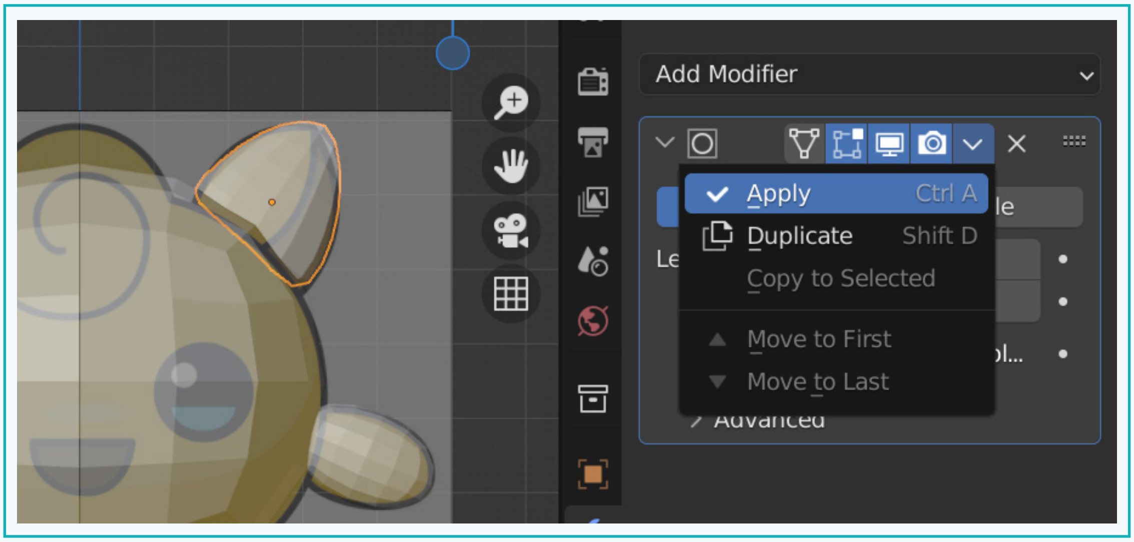
Activity

22. Manipulate the vertices to achieve the shape of the ears that matches with the reference image.

By scrolling the mouse wheel, you can change the area of influence of the transform operations.

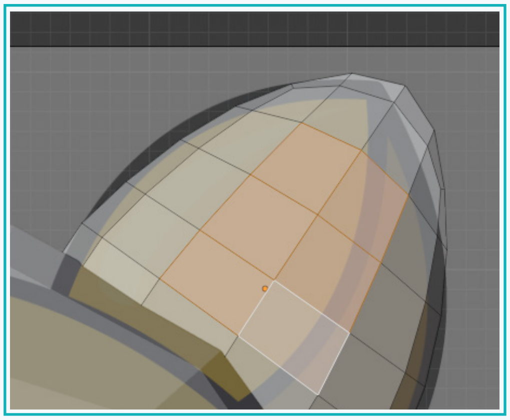


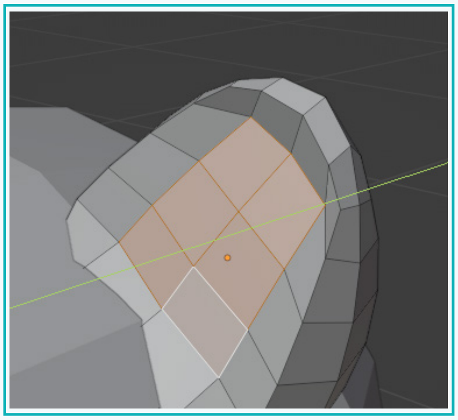
23. Add the 'Subdivision' modifier and apply it.



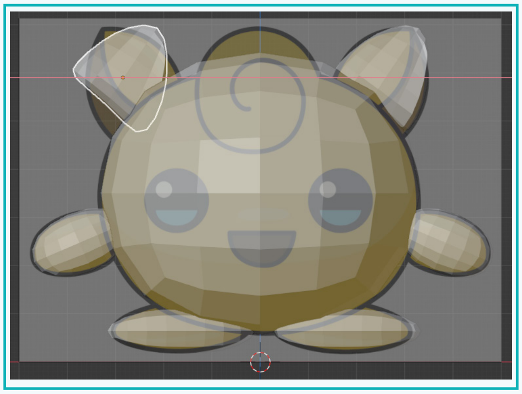
Activity

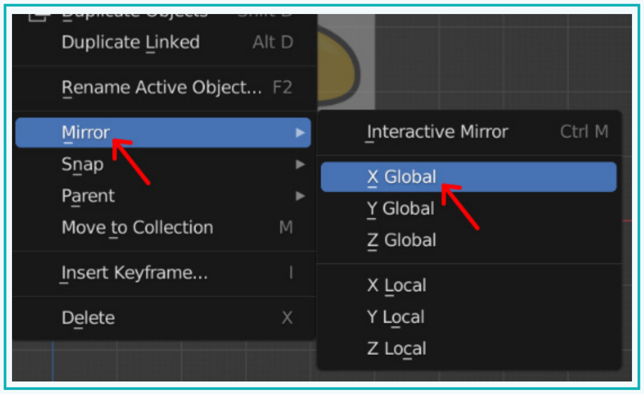
24 In order to achieve a more natural appearance for the ears, we will slightly indent the centre. Select the faces shown. To move the selected faces inward, press 'G' followed by 'Y'.

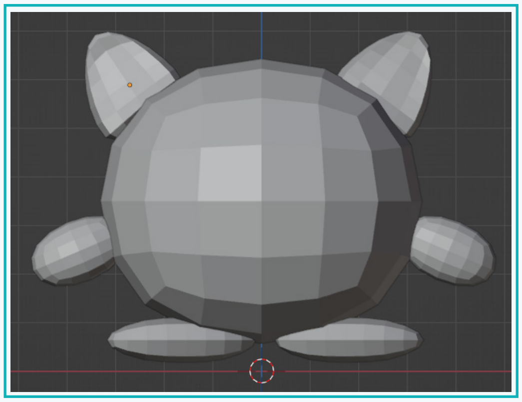




25 To create the other ear, duplicate it by pressing 'Shift + D', then move it along the x-axis and position it in the desired location. Then, 'right-click' → 'Mirror' → 'X Global'.





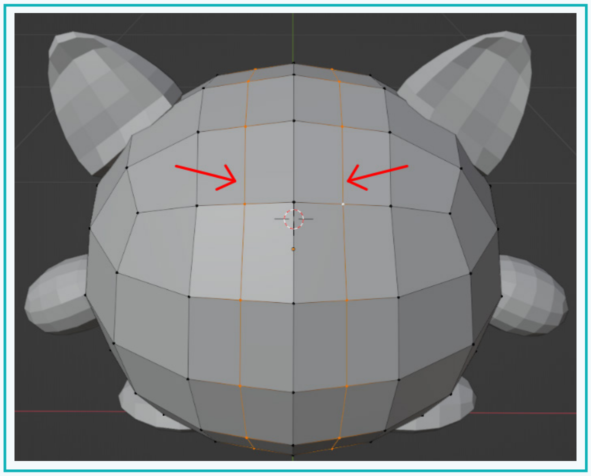


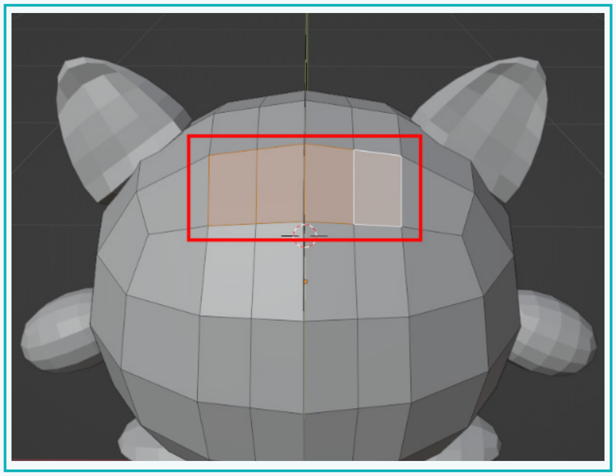
Activity

Create a Curled Tuft

Our character has a curled-up tuft of fur on its head. To create it, we need a circular arrangement of vertices that can be extruded to create the shape of the curl. For this, we need more vertices. Hence, let us begin by adding a couple of edge loops to the body of the character.

1. Press 'Ctrl + R' to add edge loops, as shown. Then, select the faces displayed.

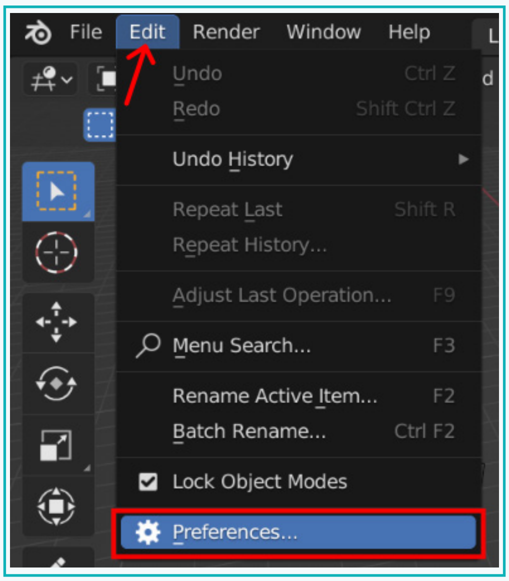


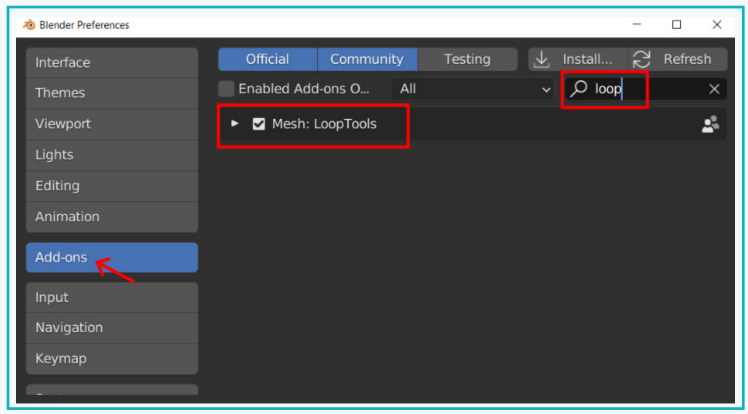


Next, we need to arrange these vertices into a circular shape. Blender provides an add-on for creating such shapes, however it is not enabled by default. In order to use it, we have to enable it.

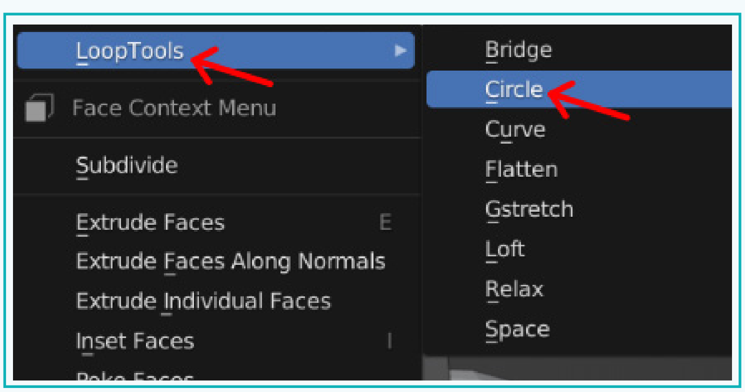
Activity

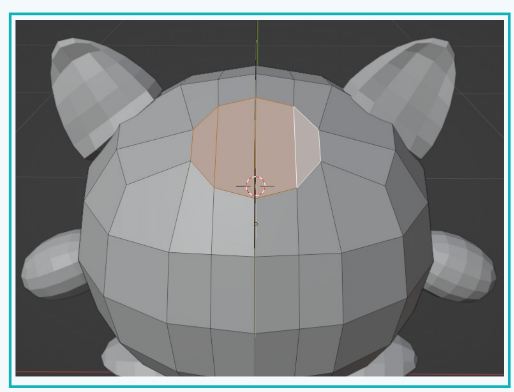
2. Click on ‘Edit’ → ‘Preferences’ → ‘Add-ons’. Type ‘LoopTools’ in the Search box and then enable it. Close the preferences window.





3. With the faces still selected, right-click → ‘LoopTools’ → ‘Circle’. This arranges the vertices present in the selection into the shape of a circle.



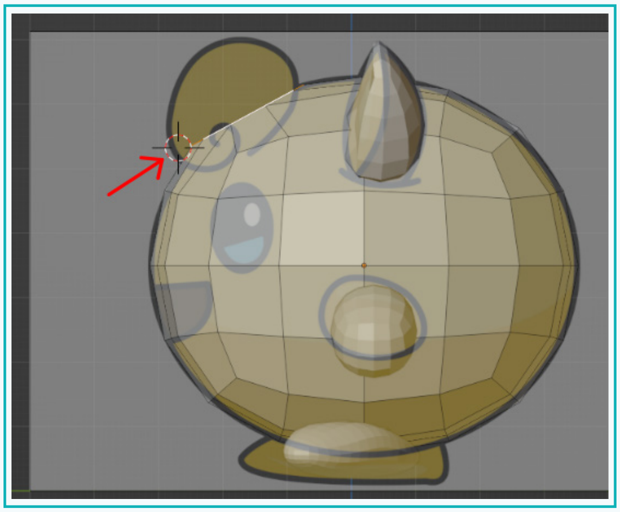


Now to create the Curl, we will use the ‘Spin’ tool in Blender.

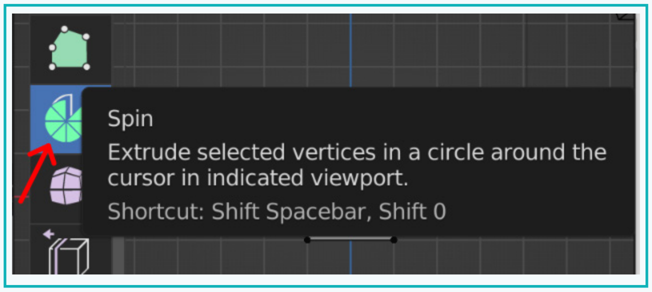
Spin Tool: It uses the position of the 3D cursor and extrudes the selected vertices/edges/faces in a circle, with the 3D cursor at the centre.

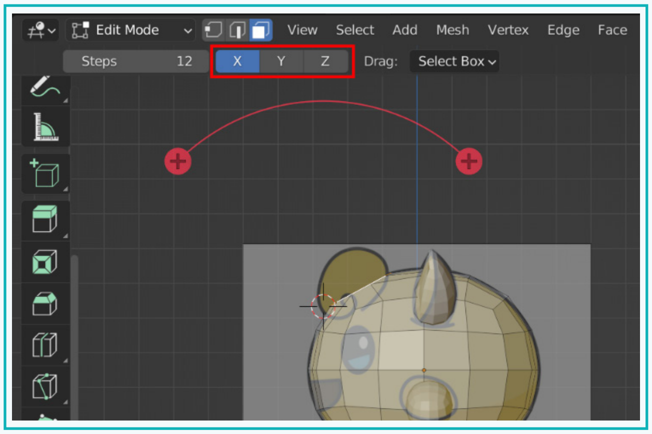
Activity

4. Go into the side view and place the 3D cursor just near the selected faces.

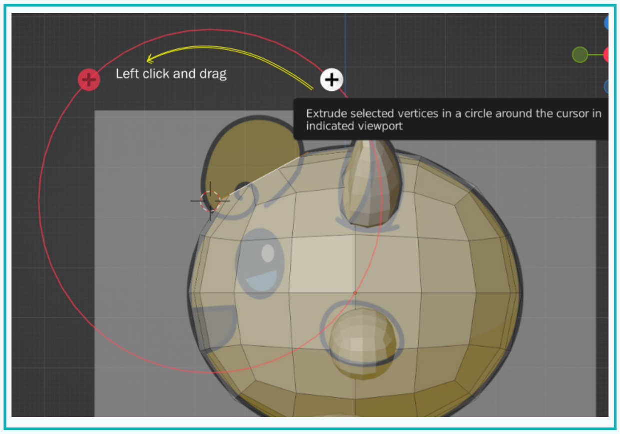


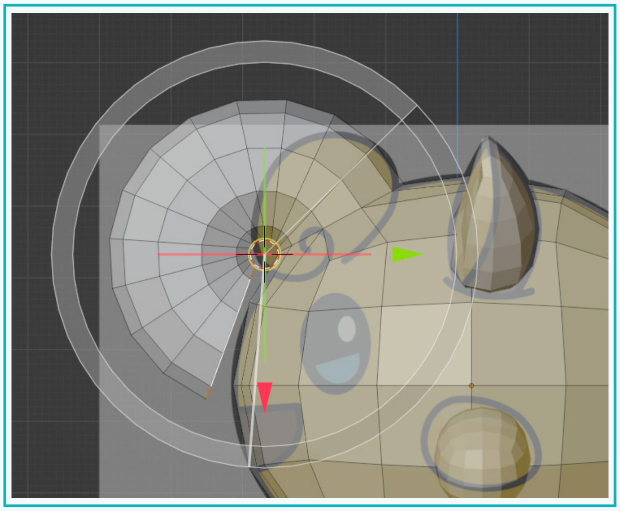
5. Click on the 'Spin' tool. Select the 'x-axis' as the axis around which the spin should occur. Click on the '+' symbol and drag in the direction the circle needs to be created.



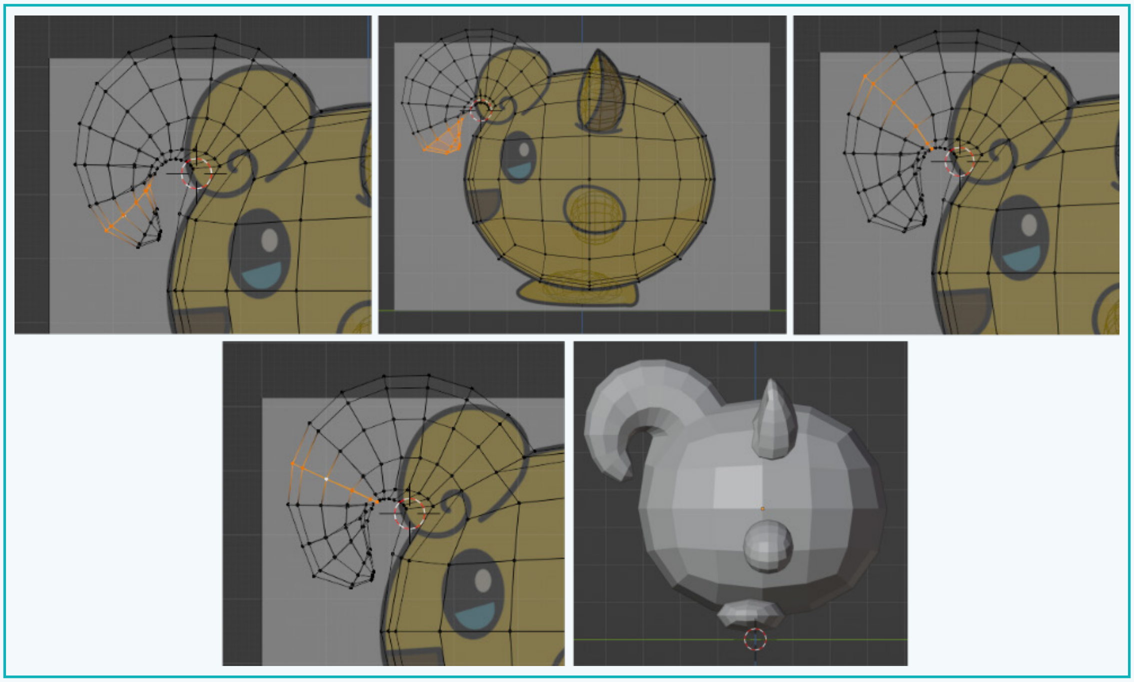


Activity



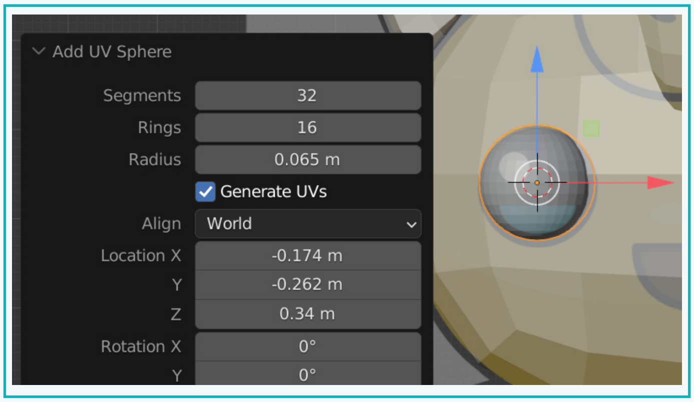


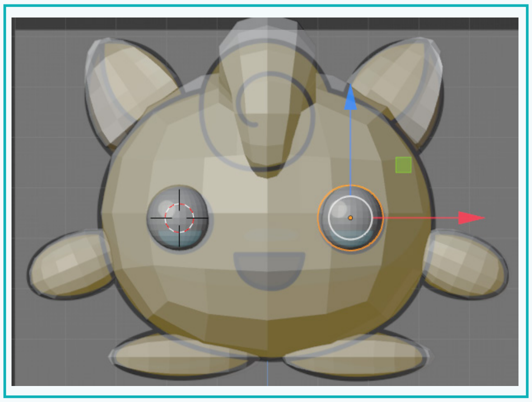
6 Scale down the end of the extruded faces and modify the individual edge loops to get a tapering effect for the curl.



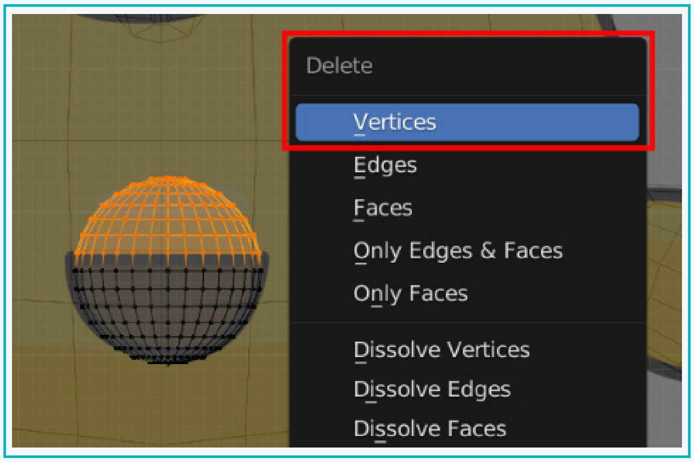
Activity

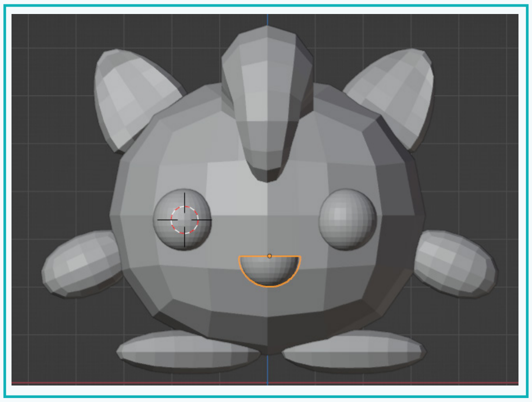
7 Create UV spheres for the eyes. Use the 3D cursor for positioning. Scale according to the reference image.





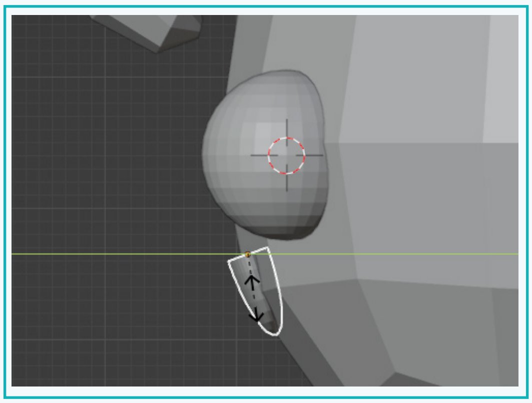
8 Also, use a UV Sphere for the mouth. To get the semicircle shape of the mouth, go into front view and select the 'Wireframe Shading' mode. Select the top half of the vertices and delete them to get the shape of the mouth.

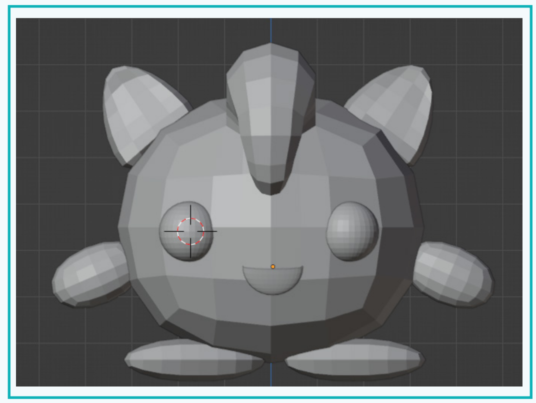




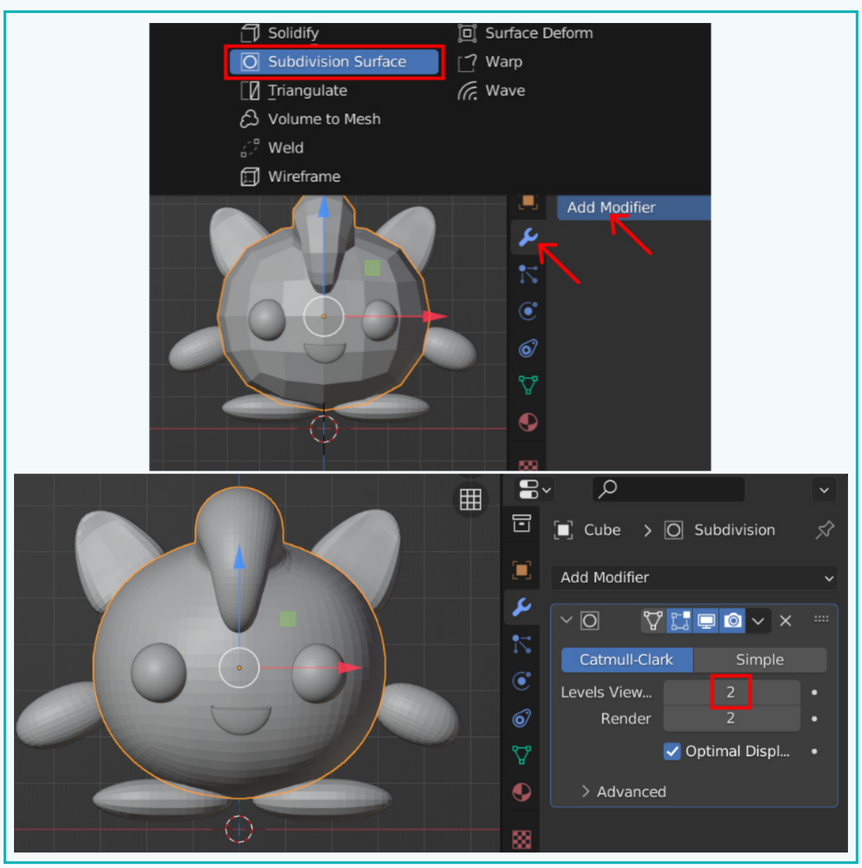
Activity

9 Go into the 'Side' view and scale the mouth on the y-axis.





10 Once all parts of the character are created as desired, apply the 'Subdivision' modifier to each part of the character to ensure smoothness for the surface of the character.



Activity

Exercise

Answer the questions with one sentence

1. Which tool did we use to accurately position objects in the 3D viewport?

▲ ____

____

2. How can we change the area of influence while using 'proportionate editing'?

▲ ____

____

State if the statements are True or False

3. The 'Loop Tools' is already enabled, and we can directly use it.

▲ ____________________

4. The 'Align to View' option aligns the image to the current view of the 3D viewport.

▲ ____________________

Fill in the blanks.

5. The 'Spin' tool takes the current position of the ____________________ and extrudes the selected element in a circle.

6. The 'Mirror' tool duplicates the selection along a particular ____________________.

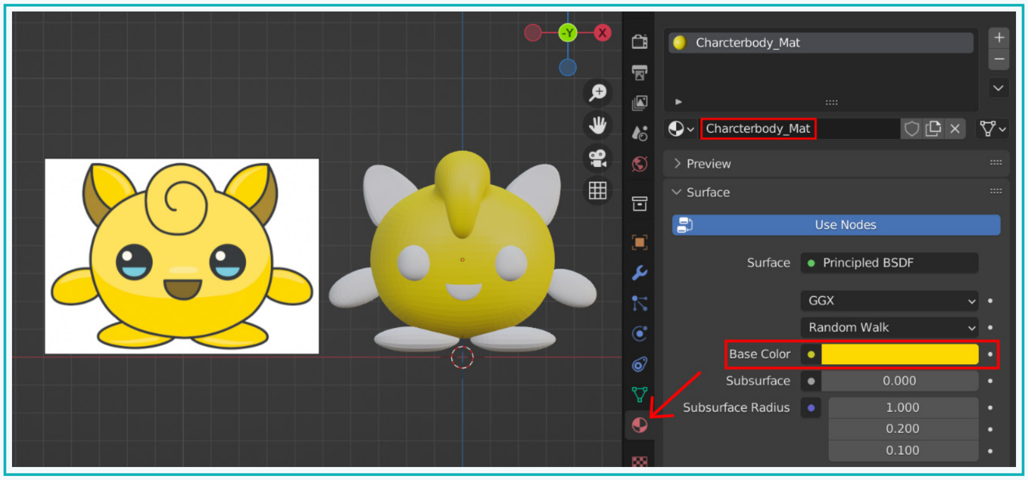
7. We used the ____________________ plugin to arrange the selected vertices in the shape of a circle.

Activity

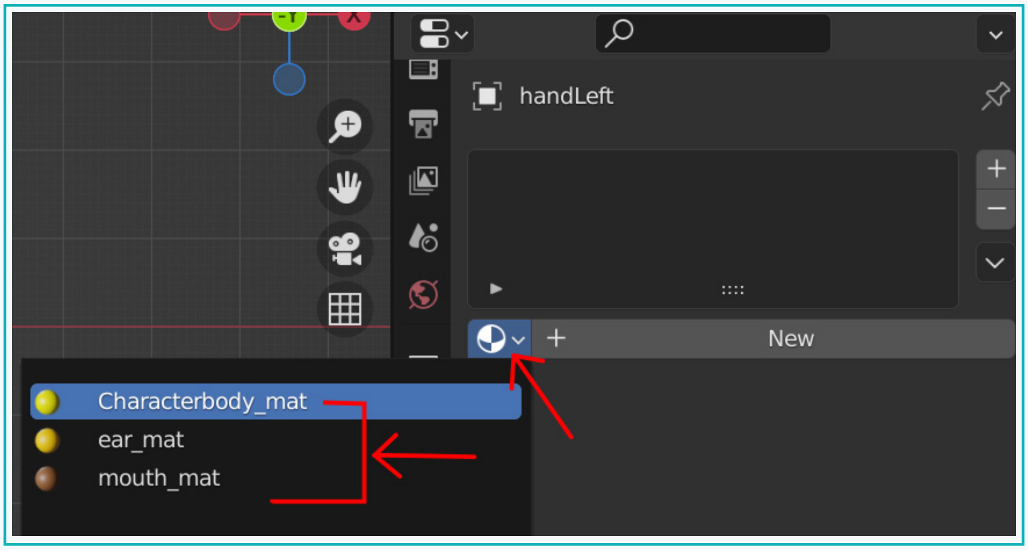
Add Materials and Textures

Now that the model of our character is ready, let us make it look attractive by adding materials and textures to it.

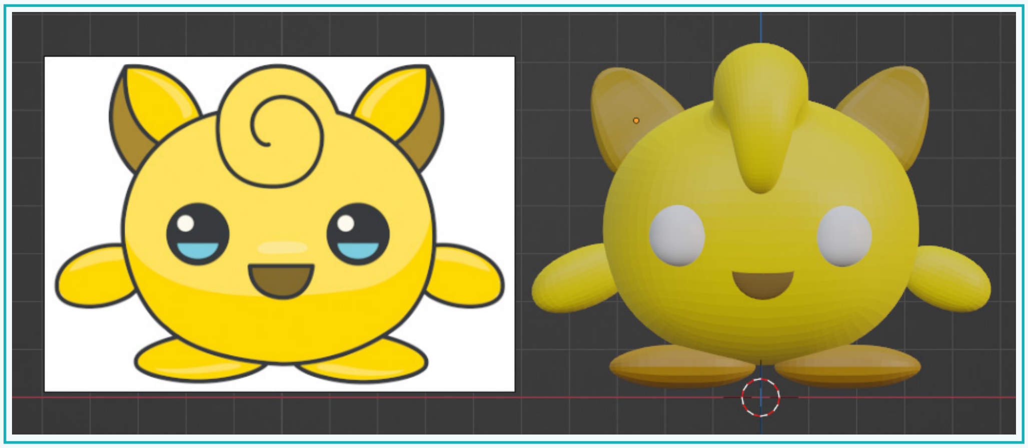
1. Select the body of the character and add a material to it. Change its ‘Base colour’ to ‘yellow’ or any colour of your choice.



2. Similarly, add materials to all the parts of the character. When applying the same colour, utilize the previously created material.



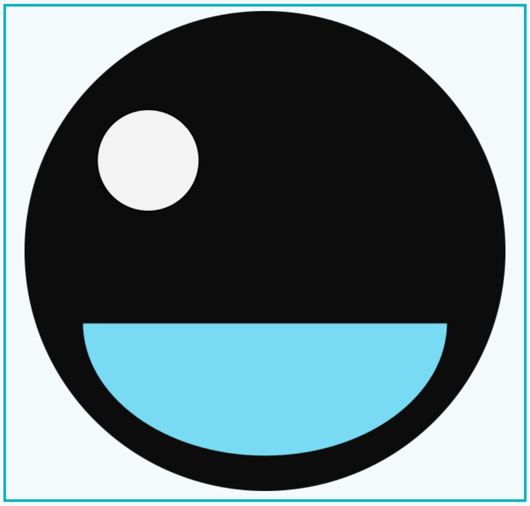
Activity



The only remaining part is the eye. The eye has a specific design, and just changing the colour of it will not produce the same look as the reference image. Our goal is to make it look as similar to the reference image, as possible.

For this, we will add a texture to the eye, i.e., we will add an image instead of a solid colour. We will use an image of an eye, as shown below.

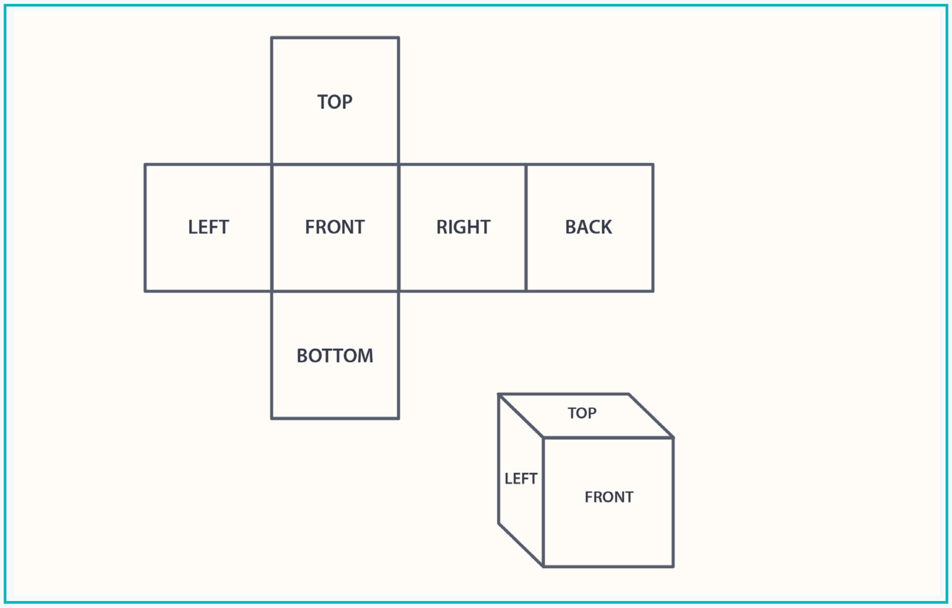
This image can be created using any image editing software such as Paint, Inkscape, and so on. We will apply this to the shape of the eye in the model. For applying a texture, we need to understand the basics of UV mapping and UV Layout. First, let us understand how UV mapping works.



Activity

UV Map

A **UV map** is a flat representation of the surface of a 3D model. A 3D model is flattened into a 2D image. Consider an example of a simple cube. If you were to flatten it out, you will have to cut it at certain seams/ edges.

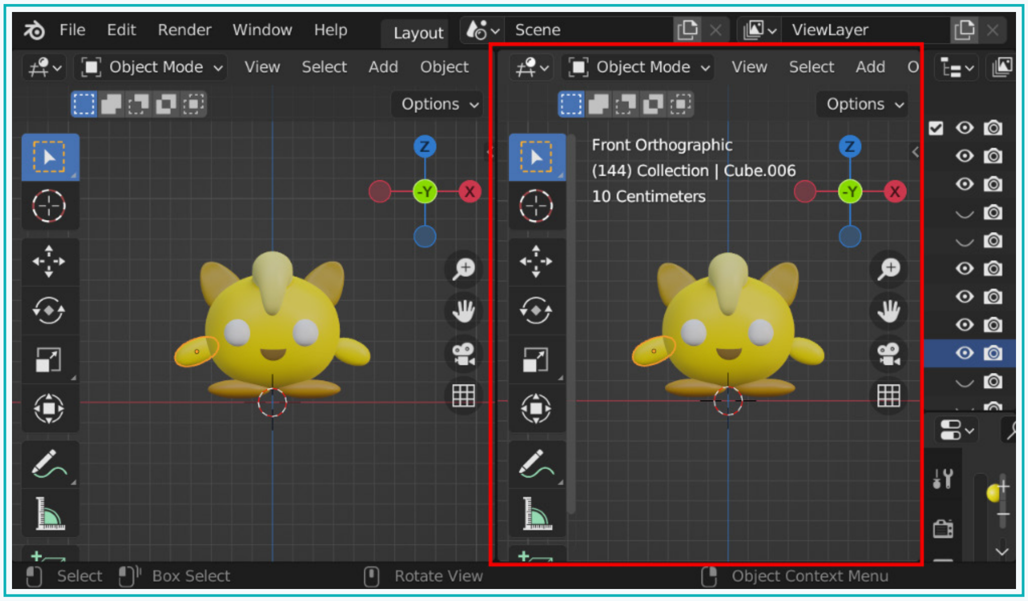


Similarly, our 3D models need to be flattened so that a custom 2D image may be positioned over our model and applied to it. This process of flattening out 3D models into 2D UV maps is called **UV Unwrapping** and in Blender, this process is done in the UV Editor.

We will perform this process of 'Unwrapping' on the eye that we have created so that we may apply the image texture to it.

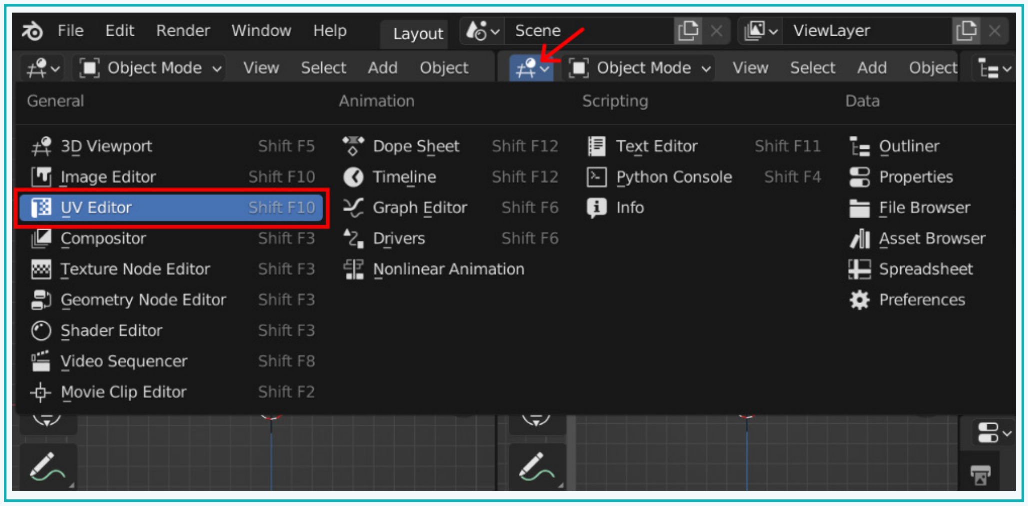
Activity

3. Open another editor window. We can do this by moving the cursor to one of the corners of any window; the cursor will transform into a plus sign. If you now drag the cursor upwards or downwards, a new window will be created horizontally, however if you drag left or right, a new window will be created vertically.

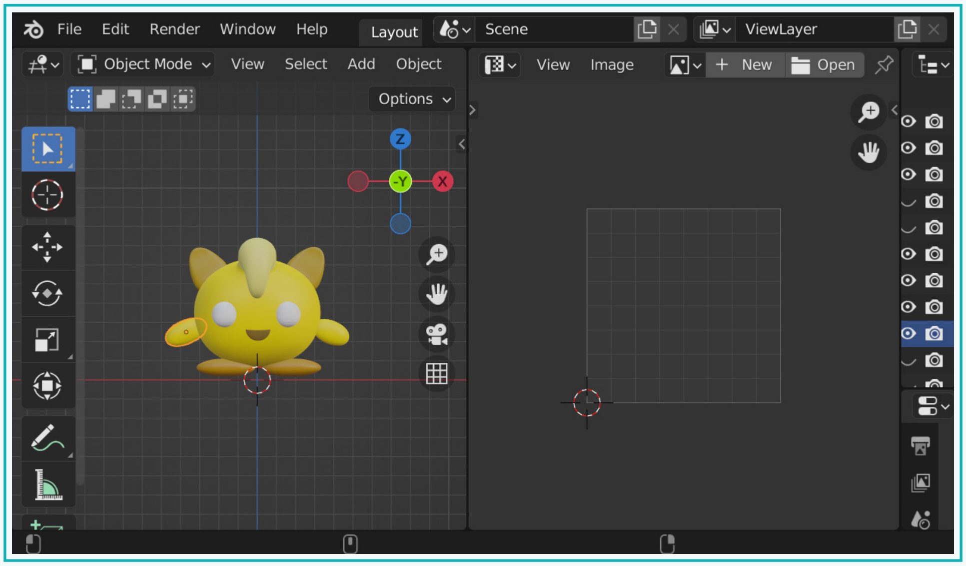


Each of these windows can be subdivided into other windows, if required using the same process.

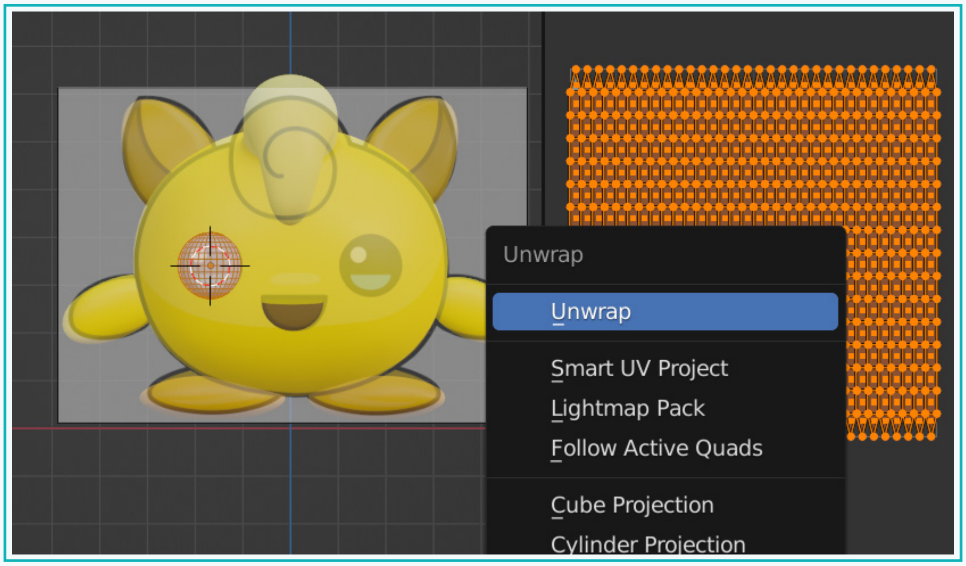
4. Change the type of window from '3D viewport' to 'UV Editor'.



Activity



5 Select the eye and tab into edit mode. Press ‘U’ for the ‘Unwrap’ menu and select ‘Unwrap’.



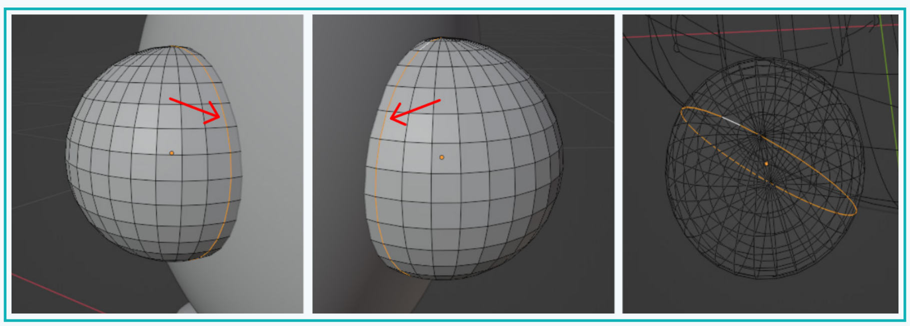
6 You will notice an error as shown below.

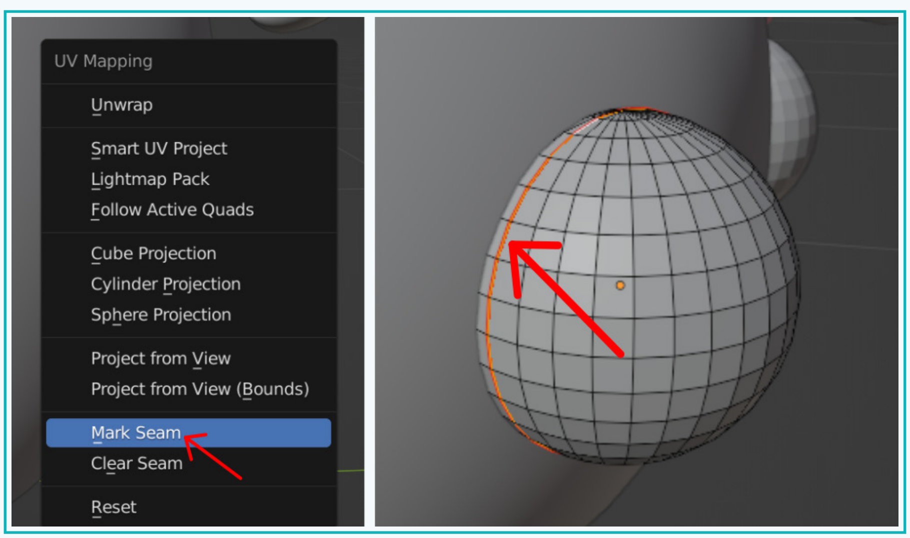
Unwrap failed to solve 1 of 1 island(s), edge seams may need to be added

Activity

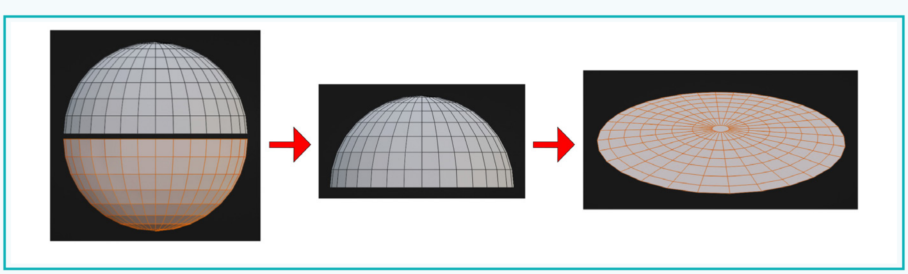
Through the error, Blender is telling us that it needs an edge seam so that it knows where to cut it out such that the 3D object can be flattened and laid flat in the UV Editor. So let us mark the seam next.

7. Select the edge loop, as shown and mark it the seam. This marks the point where the UV layout will get split.



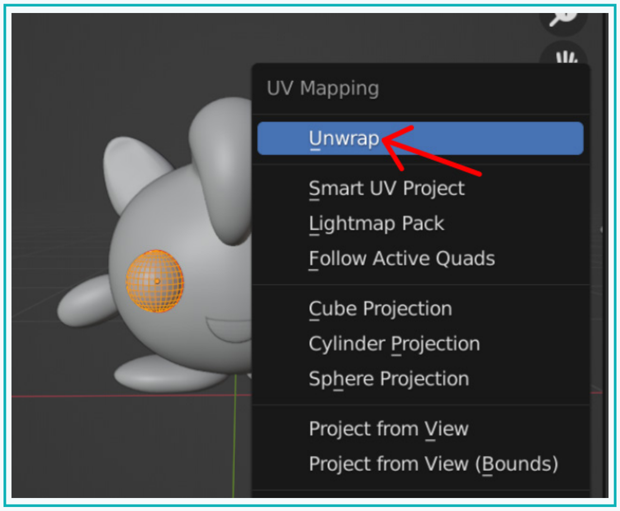


Let us understand it through the below image.

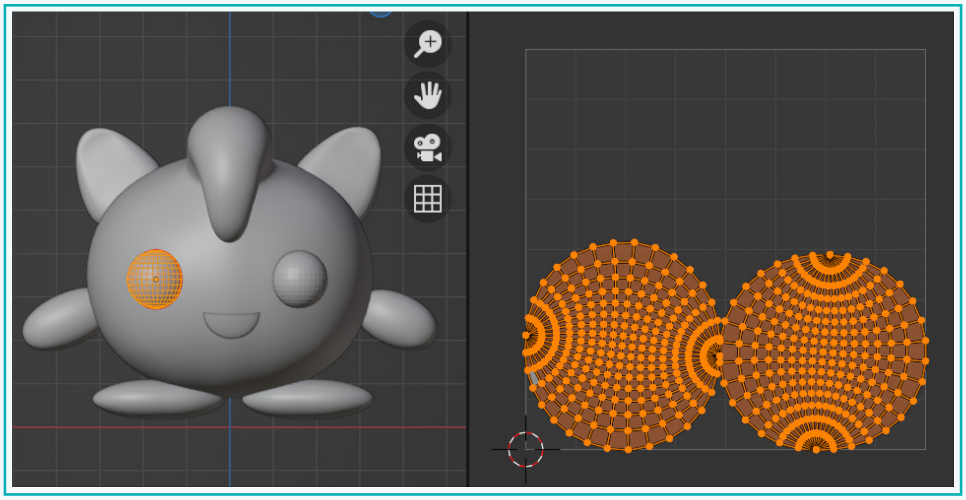


Activity

8 Next, press ‘A’ to select the faces of the eye. Press ‘U’ for the Unwrap menu. Select ‘Unwrap’.

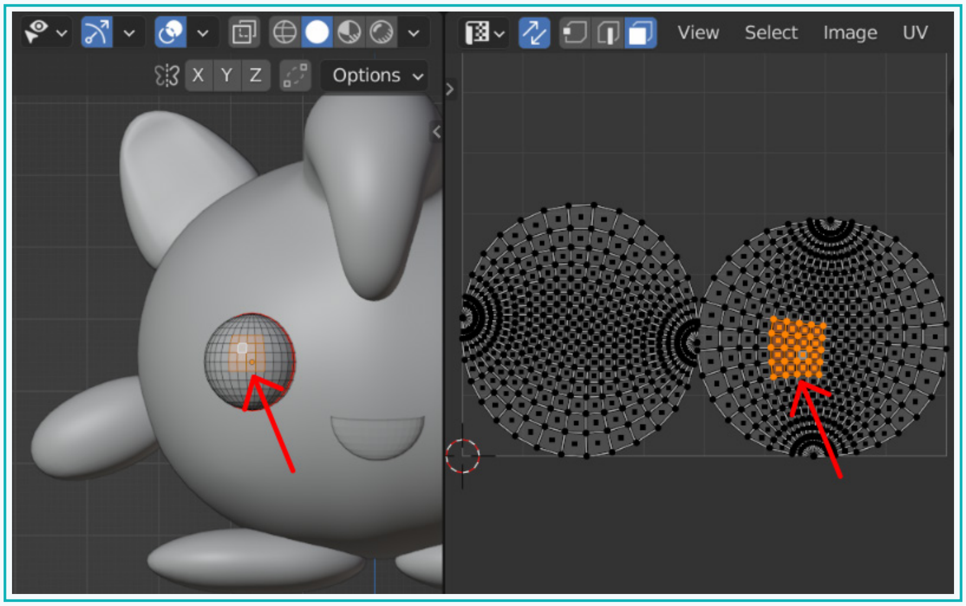


9 The shape of the eye gets unwrapped into 2 circles. The spherical shape of the eye gets split up at the seam that we had marked in the previous step.



Activity

10 With the ‘UV Sync Selection’ on, select any face on one of the circles. By doing this, you can figure out which one of the 2 circles depicts the front half of the eye that is visible to us.



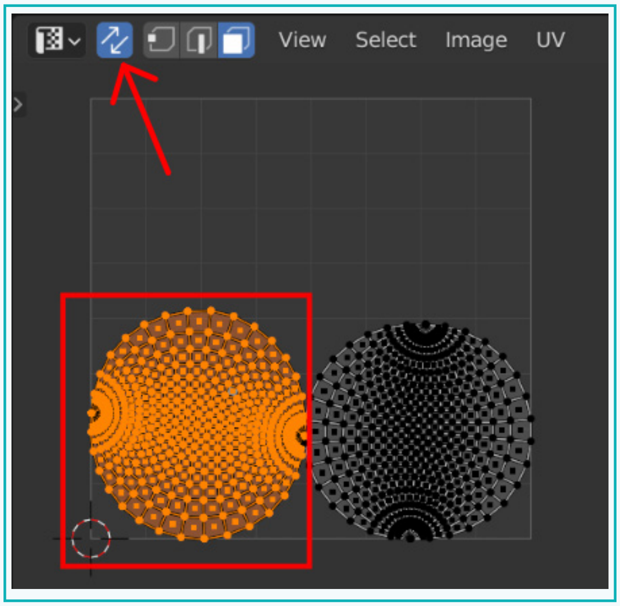
As we can see, the faces in the right circle represent the front half of the eye. We will apply the texture only to the front half of the eye. For this, we need to keep the UV layout of only the front half in the UV Layout area. Hence, we move the part that is not required out of the UV Layout area.

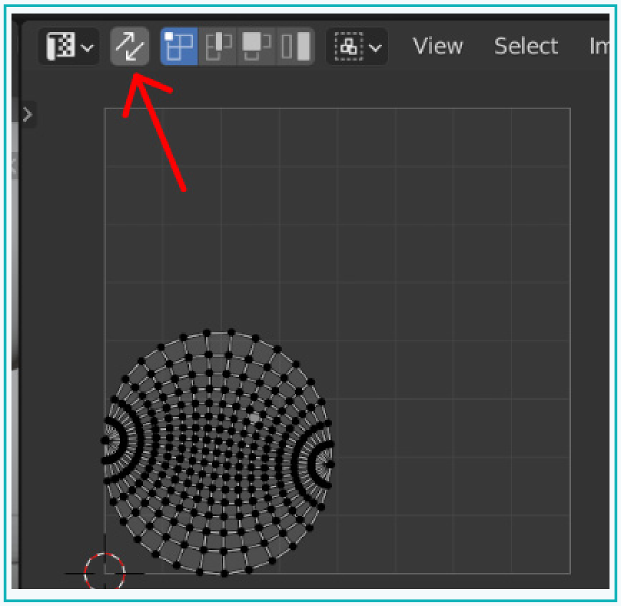
We will use the ‘UV Sync Selection’ option for this purpose. Enabling the ‘Sync Selection’ button ensures that the selection of components in the 3D View is synchronized with their corresponding elements in the UV/Image editor. When the button is turned off, only the selected faces are displayed in the UV/Image editor.

11 With ‘UV Sync Selection’ on, select all the faces of the circle representing the back half of the eye. You can do this by selecting any one face and then, pressing the ‘L’ key to select all the linked faces.

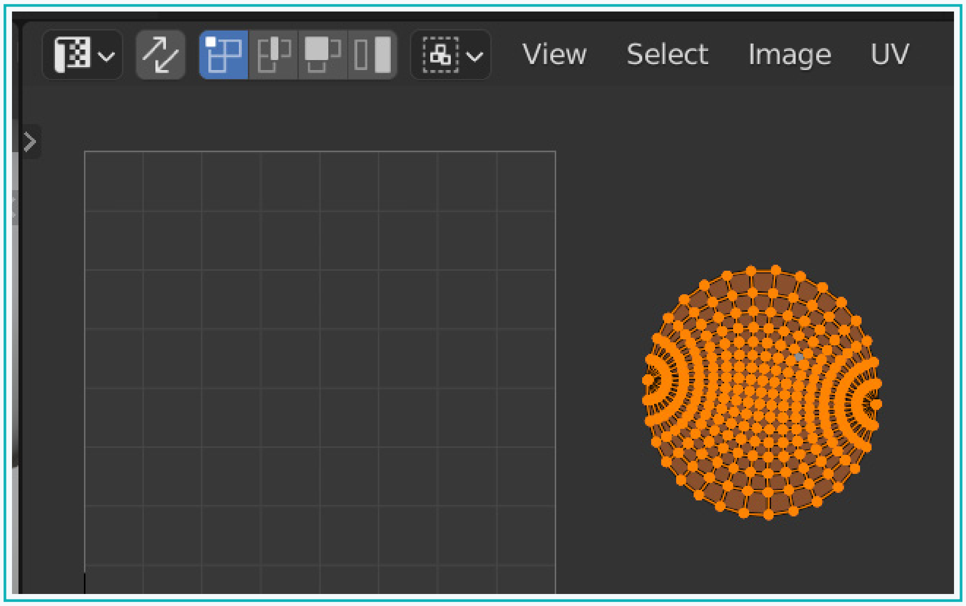
Activity

These selected faces need to be moved out of the UV Layout area as we do not want to apply texture to it. With the faces selected, switch off the 'UV Sync Selection' mode. This will show only the selected face(the back half of the sphere) in the UV editor.



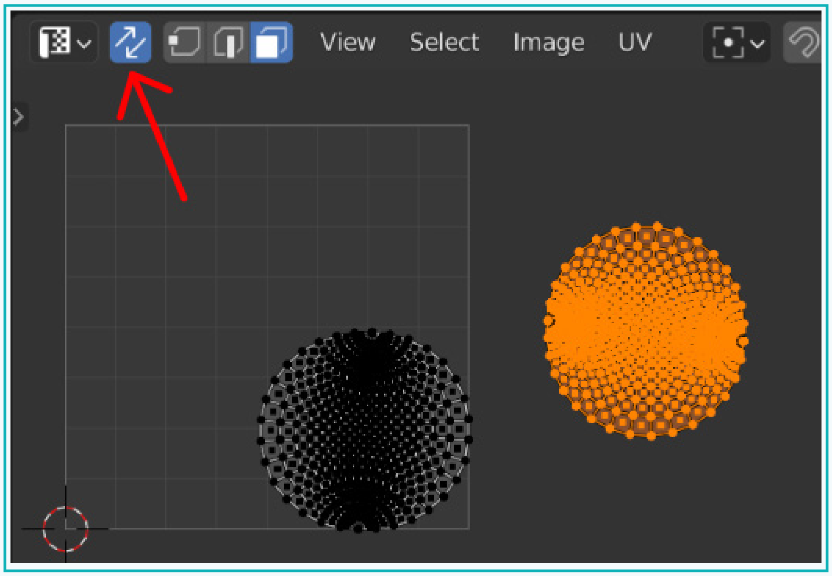


12 Press 'A' to select all the faces and press 'G' to move them out of the UV display area.

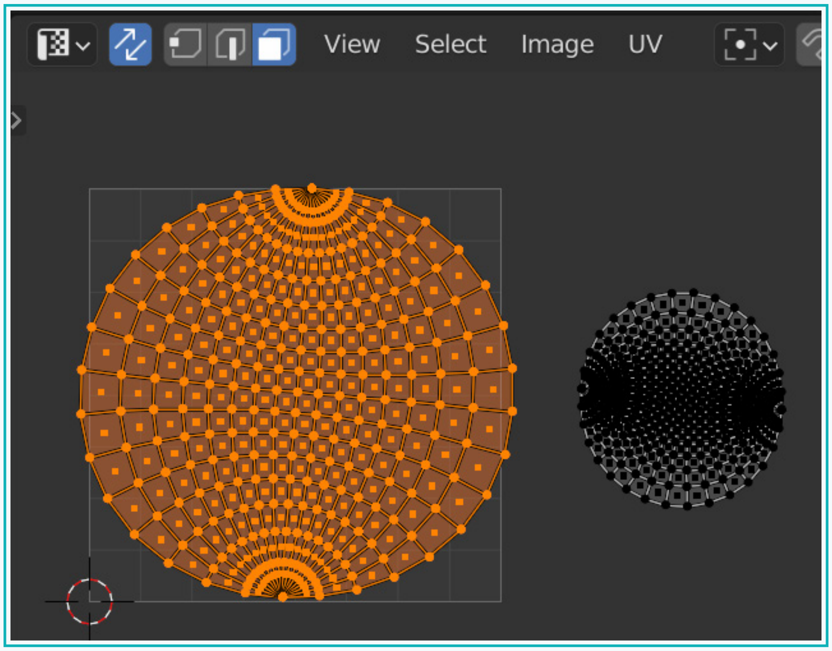


Activity

13. Now switch on ‘UV Sync Selection’ again. This will reveal the UV map of the front half.

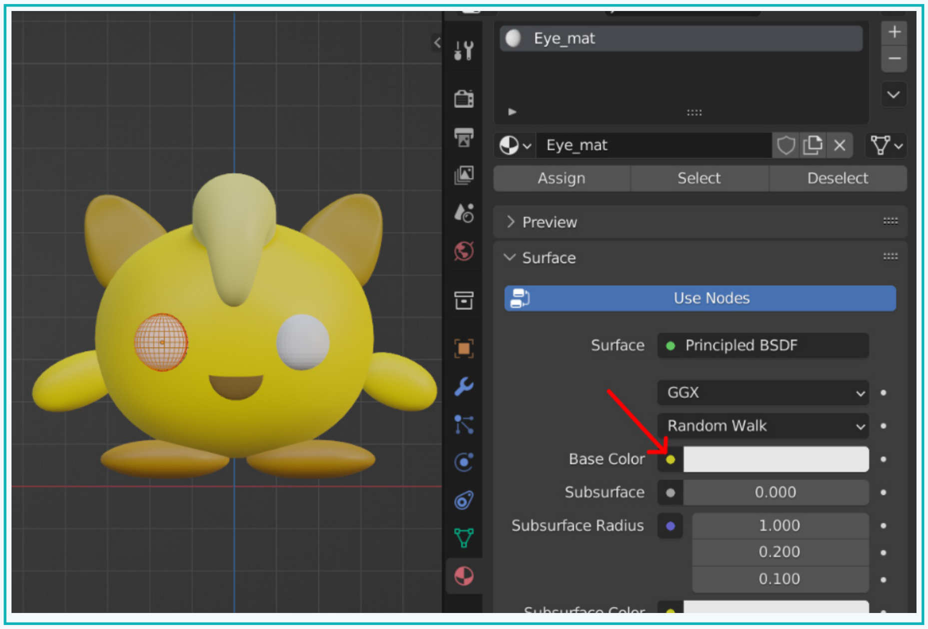


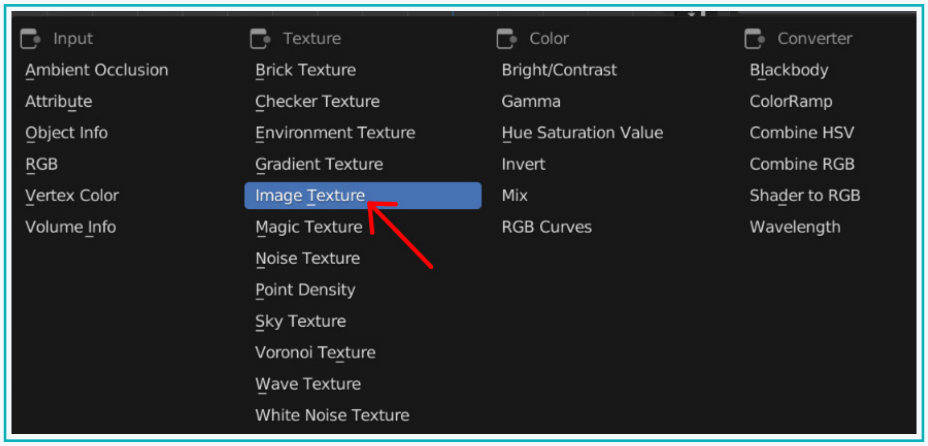
14. Using **Box Select or Linked Selection,** select all the faces of the front half and scale it up to cover the UV Layout area.



Activity

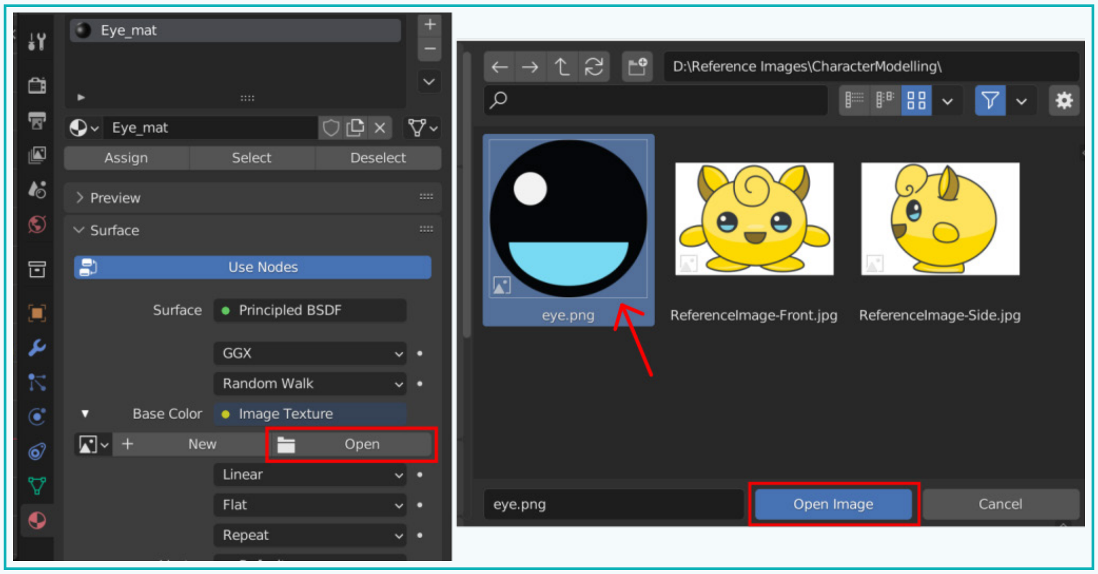
15 Now, add material to the eye. Click on the dot next to the Base colour and select 'Image Texture'.



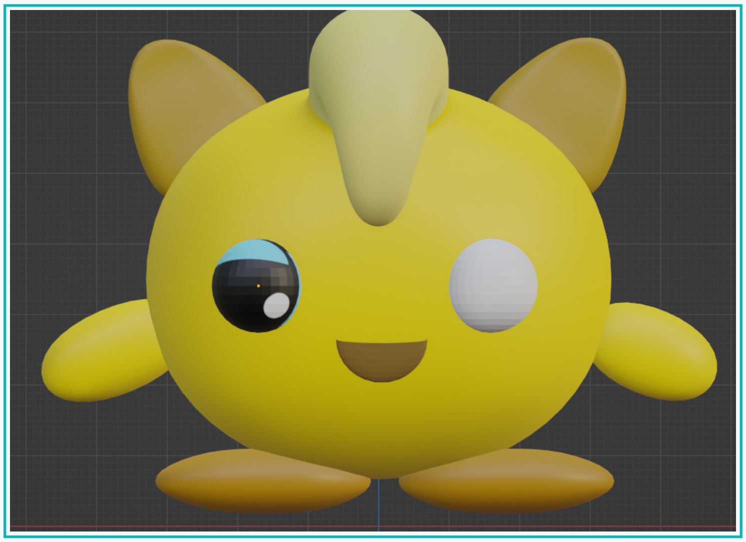


Activity

16 Click on ‘Open’ and navigate to the folder where the eye image is saved.

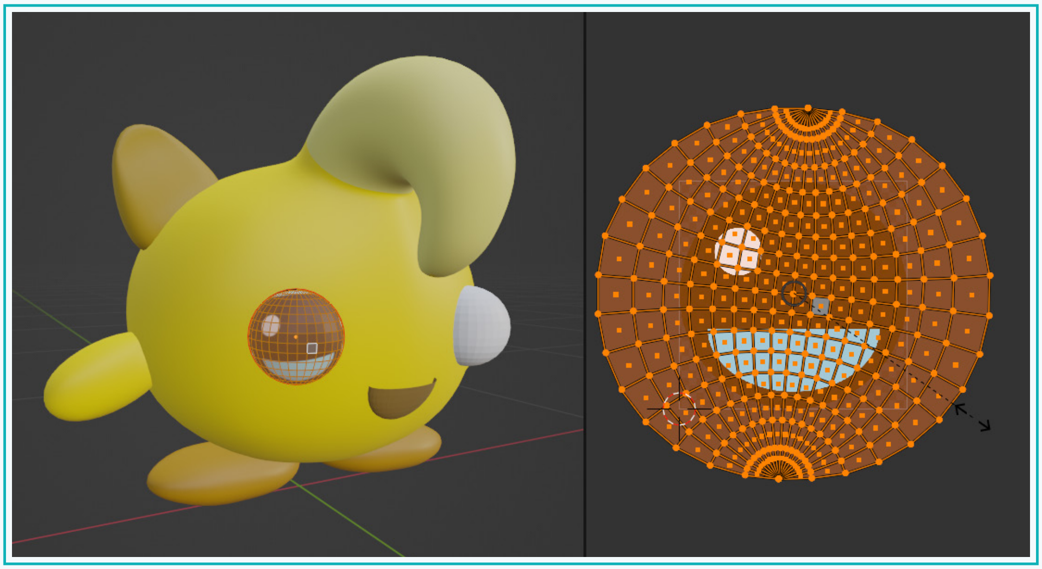


17 The image gets projected onto the eye. However, it is not aligned correctly.

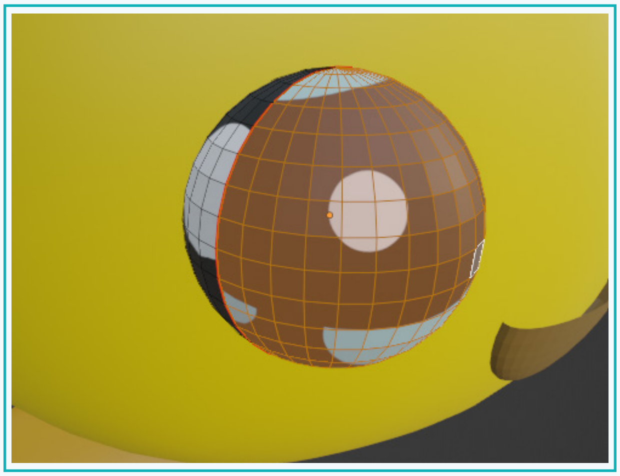


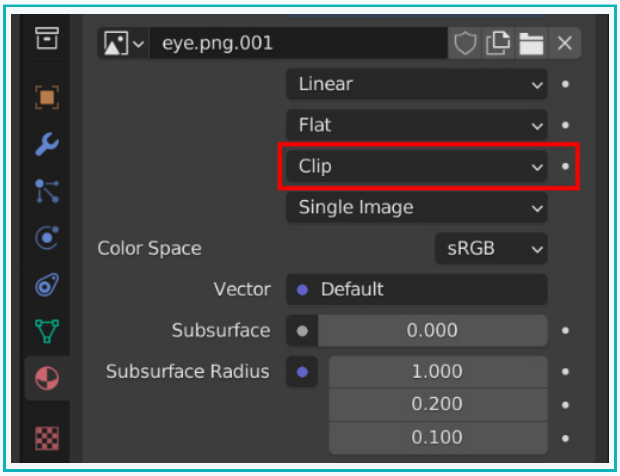
Activity

18 To align correctly, select the faces in the UV Layout. Then adjust the positioning using 'R' for rotating, 'S' for scaling, and 'G' for moving.



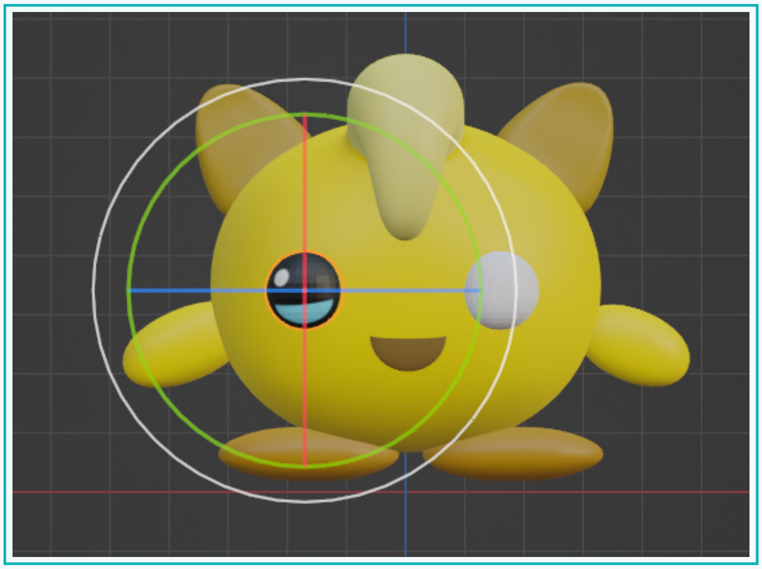
19 On scaling, you will notice that the eye pattern gets repeated. This can be avoided by changing the 'Repeat' property to 'Clip' in the Material properties window.





Activity

20 Rotate the UV sphere in the 3D viewport, if required, to get the desired look for the eye.



21 This can now be duplicated to create the other eye.

The 3D model of our character is ready.

In this lesson, we created a 3D model of a character. We learnt how to use the Spin tool, and applied materials to our model. We learnt the basics of UV Layout and UV Unwrapping and applied textures to make the character look realistic.

Exercise

Answer the questions with one sentence

8. What is a UV map?

▲ ______________________________

____

9. What is a seam?

▲ ______________________________

____

10. In which editor is the process of UV Unwrapping performed?

▲ ______________________________

____

State if the statements are True or False

11. It is not possible to reuse previously created material.

▲ ______________

12. An image texture allows us to add an image instead of a solid colour to an object.

▲ ______________

Exercise

Fill in the blanks.

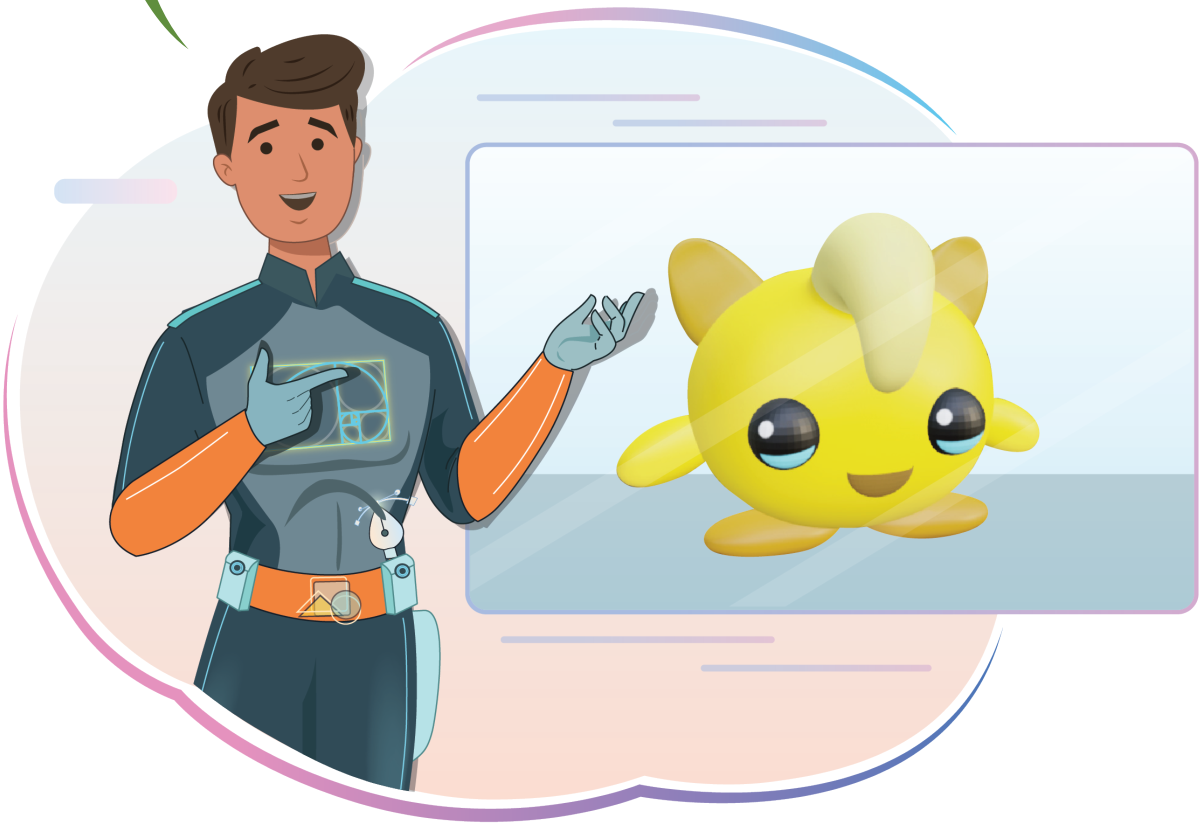
13. The ____________________ option ensures that the selection of components in the 3D View is synchronized with their corresponding elements in the UV/Image editor.

14. The ‘L’ key allows one to select all ____________________ vertices/edges/faces.

15. The process of flattening out 3D models into 2D UV maps is called ____________________.

Hope you enjoyed the journey of character modelling. You can now create more exciting characters for your game.

Happy modelling!



Tech Flash

- **UV Unwrapping** is the process of flattening out 3D models into 2D UV maps.
- **UV Sync Selection** option ensures that the selection of components in the 3D View is synchronized with their corresponding elements in the UV/Image editor.
- **UV map** is a 2D representation of a 3D object.
- **A seam** is where Blender cuts a 3D object and lays it flat in the UV Editor.